An Explainable Deep Learning Framework for Speech-Based Depression Detection

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DECLARATION

I declare that this thesis is my own work and this thesis/dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. I retain the right to use this content in whole or part in the future works (such as articles or books).

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Name of Supervisor:

Signature of the Supervisor: Date:

DEDICATION

I dedicate this thesis to my beloved parents, whose unconditional love, endless encouragement, and unwavering belief in me have been my greatest source of strength throughout my academic journey. Their sacrifices and support have made this achievement possible.

I also dedicate this work to my respected supervisor, Prof. Subha Fernando, and my co-supervisor, Dr. Visula, for their invaluable guidance, continuous encouragement, constructive feedback, and unwavering support throughout this research. Their expertise, patience, and mentorship have been instrumental in the successful completion of this thesis.

Finally, I dedicate this thesis to my family, friends, lecturers, and everyone who supported and believed in me throughout this journey. Their encouragement and kindness inspired me to persevere and achieve this important milestone.

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ABSTRACT

Depression is a major mental health disorder that often remains underdiagnosed due to the subjective nature of traditional clinical assessments. Current diagnostic methods rely heavily on interviews and self-reported symptoms, which can be inconsistent and time-consuming. In contrast, human speech provides an objective and non-invasive source of information, as acoustic features such as pitch variation, energy levels, and speech patterns naturally reflect emotional and psychological states. Recent advancements in artificial intelligence have enabled the use of deep learning techniques to analyze speech data for depression detection with promising accuracy.

However, existing deep learning models face a significant limitation: they operate as "black box" systems, providing predictions without explaining the reasoning behind them. This lack of interpretability creates a critical gap between technological capability and clinical usability. Healthcare professionals require transparent and explainable results to validate decisions and ensure patient safety. Without such explanations, even highly accurate models are difficult to trust and adopt in real-world clinical settings.

To address this challenge, this research proposes an explainable deep learning framework for speech-based depression detection that integrates predictive accuracy with interpretability. The hypothesis of this study is that incorporating explainability into deep learning models can improve transparency and clinical trust while maintaining strong classification performance. The system takes speech recordings as input and processes them through a structured pipeline, where the audio data is transformed into representations that capture both temporal and frequency-based characteristics. A hybrid deep learning model is then used to classify the input as depressed or non-depressed, producing a prediction along with a confidence score.

An explainability component is incorporated to provide insights into the model's decision-making process. This component highlights the most influential segments of the speech signal and identifies key acoustic features that contribute to the prediction. As a result, the system outputs not only the classification result but also visual evidence that supports the decision, making it more understandable for clinicians.

From a theoretical perspective, this research contributes to the field of artificial intelligence by addressing the balance between model performance and interpretability in sensitive healthcare applications. It extends existing work by emphasizing transparency as a core requirement in model design rather than an optional feature. The framework will be evaluated using a balanced dataset, assessing both classification performance and the relevance of generated explanations.

In conclusion, this study presents a novel approach to speech-based depression detection by combining deep learning with explainability. The proposed framework enhances trust and usability by providing interpretable predictions, thereby supporting

more reliable and responsible application of artificial intelligence in mental healthcare.

CHAPTER 1
INTRODUCTION

1.1 Introduction

Depression is one of the most prevalent mental health disorders worldwide, significantly affecting individuals' emotional, cognitive, and social functioning. Early and accurate detection is crucial for effective intervention and treatment. However, traditional diagnostic approaches primarily rely on subjective clinical assessments, including interviews and self-reported questionnaires, which may lead to inconsistencies and delayed diagnosis.

Recent advancements in artificial intelligence have enabled the development of automated systems capable of analyzing behavioral signals, such as speech, to support mental health assessment. Human speech contains valuable acoustic and temporal characteristics, including pitch variation, speech rate, and energy distribution, which can reflect underlying psychological states [1]. Deep learning techniques have shown promising results in extracting such patterns directly from speech data, achieving high performance in depression detection tasks [2].

Despite these advancements, a critical limitation persists: most deep learning models function as opaque systems that do not provide insight into their decision-making processes. This lack of interpretability restricts their applicability in clinical settings, where transparency and justification are essential. Therefore, there is a growing need to develop intelligent systems that not only perform accurate predictions but also provide understandable explanations to support clinical decision-making.

1.2 Objectives

The primary objectives of this research are:

- To conduct a critical literature review of existing approaches for speech-based depression detection and identify their limitations.
- To perform an in-depth study of relevant technologies, including deep learning and explainable artificial intelligence, and define existing research gaps.
- To design and develop an explainable deep learning framework that integrates prediction and interpretability for depression detection.
- To develop an extension that bridges the gap between high model accuracy and lack of transparency in existing systems.
- To evaluate the effectiveness of the proposed framework in terms of both predictive performance and interpretability.
- To assess the contribution of the proposed approach to the broader field of artificial intelligence, particularly in healthcare applications.

1.3 Problem in Brief

While deep learning models have demonstrated strong performance in detecting depression from speech data, they suffer from a fundamental limitation: lack of interpretability[3]. These models operate as “black boxes,” providing predictions without explaining the underlying reasoning.

This creates a critical challenge in healthcare applications. Clinicians require clear and interpretable evidence to validate any diagnostic support provided by artificial intelligence systems. Without such transparency, the reliability and trustworthiness of these models remain questionable[1]. As a result, there is a significant gap between the technical capabilities of existing systems and their practical usability in clinical environments.

1.4 Background and Motivation

Speech-based depression detection has gained increasing attention due to its non-invasive and cost-effective nature. Research has shown that individuals experiencing depression often exhibit distinct speech characteristics, such as reduced pitch variability, slower speech rate, and increased pauses [3]. Machine learning and deep learning models have been applied to analyze these features, with convolutional and recurrent neural networks achieving state-of-the-art results [4].

However, the majority of these models focus primarily on improving classification accuracy while neglecting interpretability. In sensitive domains such as healthcare, this limitation becomes a major barrier to adoption. Clinicians must understand why a model produces a particular outcome before integrating it into decision-making processes.

The motivation for this research stems from the need to bridge this gap by developing a system that combines predictive accuracy with explainability. By providing transparent insights into model behavior, the proposed approach aims to enhance trust and facilitate the practical deployment of artificial intelligence in mental health diagnostics.

1.5 Novel Approach to Speech-Based Depression Detection

This research proposes an explainable deep learning framework that integrates classification and interpretability into a unified system. Unlike traditional approaches that focus solely on prediction, the proposed method emphasizes transparency as a core component[3].

The system processes speech recordings by converting them into representations that capture both temporal and frequency-based characteristics. A hybrid deep learning architecture is employed to learn complex patterns associated with depressive speech. The key novelty lies in the integration of an explainability layer that provides visual and feature-level insights into the model's decision-making process.

This extension bridges the identified research gap by enabling the model to not only predict depression but also explain the reasoning behind its predictions. As a result, the system becomes more suitable for clinical applications where interpretability is essential.

1.6 Resource Requirements

The successful implementation of this research requires the following resources:

- **Hardware Resources:**
    - A development machine with sufficient storage capacity
    - Access to graphical processing units for efficient model training
- **Software Resources:**
    - Programming language: Python
    - Deep learning frameworks such as TensorFlow or PyTorch
    - Audio processing libraries for feature extraction
    - Tools for data analysis and visualization
- **Data Resources:**
    - Access to speech datasets containing recordings from both depressed and non-depressed individuals
    - Preprocessing tools for cleaning and preparing audio data

1.7 Structure of the Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2** presents a detailed literature review of existing methods and technologies related to speech-based depression detection and explainable artificial intelligence.
- **Chapter 3** describes the methodology, including data processing techniques, model architecture, and system design.
- **Chapter 4** outlines the implementation details and experimental setup.
- **Chapter 5** presents the results and evaluation of the proposed framework.
- **Chapter 6** discusses the findings, limitations, and contributions of the research.
- **Chapter 7** concludes the study and summarizes key outcomes.

1.8 Summary

This chapter introduced the research problem, highlighting the limitations of existing deep learning models in speech-based depression detection, particularly the lack of interpretability. It outlined the objectives of the study and emphasized the importance of bridging the gap between model performance and transparency. The proposed approach focuses on developing an explainable deep learning framework that enhances both accuracy and clinical usability. The chapter also provided an overview of the required resources and the structure of the thesis, establishing a foundation for the subsequent chapters.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Following the problem identified in Chapter 1, this chapter critically reviews existing research on speech-based depression detection and the role of artificial intelligence in mental health analysis. While prior studies have demonstrated promising performance using machine learning and deep learning techniques, significant challenges remain, particularly in terms of interpretability and clinical applicability.

This chapter examines the evolution of approaches in this domain, analyzes their strengths and limitations, and identifies key technological gaps. The discussion is structured thematically, covering traditional machine learning, deep learning methods, and emerging explainable artificial intelligence approaches. The chapter concludes by defining the research gap that motivates the proposed solution[5].

2.2 Chronological Review of Prior Work

Research on automated depression detection from speech has evolved from handcrafted acoustic analysis to deep learning, and more recently toward explainable and clinically usable systems. This section traces that progression and identifies persistent challenges that motivate the present study.

2.2.1 Early development of Speech-Based Depression Detection

The earliest work in this area emerged from clinical psychiatry and speech science, long before modern machine learning. Clinicians observed that depressed patients often exhibit psychomotor retardation, reflected in slower speech, longer pauses, reduced vocal energy, and flatter intonation (Monotone pitch, reduced prosody). These observations established speech as a potential behavioral biomarker for mood disorders.

In the 1990s and early 2000s, researchers began quantifying these observations using handcrafted acoustic features. Studies extracted pitch (fundamental frequency), energy, speech rate, pause duration, jitter, shimmer, and spectral measures from recorded interviews. Classifiers such as Support Vector Machines (SVM), Gaussian Mixture Models (GMM), and Hidden Markov Models (HMM) were applied to distinguish depressed from non-depressed speakers. Feature sets often included MFCCs, formant frequencies, and zero-crossing rate.

A major limitation of this era was small, heterogeneous datasets and inconsistent labeling criteria. Labels were derived from clinical interviews, self-report scales (e.g., BDI, PHQ), or DSM-based diagnosis, making cross-study comparison difficult. Nevertheless, these studies established a consistent finding: paralinguistic cues—how something is said—carry diagnostic information beyond lexical content.

The introduction of standardized corpora marked an important milestone. The DAIC-WOZ (Distress Analysis Interview Corpus — Wizard-of-Oz) dataset, developed for the AVEC (Audio/Visual Emotion Challenge) depression sub-challenge, provided clinical interview recordings with PHQ-8–validated labels. This enabled reproducible benchmarking and shifted the field toward data-driven methods. Early DAIC-WOZ baselines relied on engineered features (e.g., COVAREP, openSMILE) combined with traditional classifiers, achieving moderate but promising results and confirming that speech alone contains depression-related signal.

During this period, evaluation practices were often segment-level and sometimes ignored speaker independence, leading to optimistic performance estimates when the same participant appeared in both training and test sets. Despite these methodological weaknesses, early work firmly established the feasibility of speech-based depression screening as a research direction.

2.2.2 Recent developments and Future trends

Recent years have seen a shift from handcrafted features to deep learning, particularly architectures that operate directly on time-frequency representations of speech.

Convolutional Neural Networks (CNNs) applied to mel spectrograms or log-mel features have become common, treating spectrograms as image-like inputs where local time–frequency patterns can be learned automatically. Recurrent Neural Networks (RNNs) and LSTM/GRU models capture longer temporal dependencies in frame-level or segment-level feature sequences. Hybrid CNN–LSTM architectures combine spatial and sequential modeling. More recently, self-supervised pre-trained models (e.g., wav2vec 2.0, HuBERT) fine-tuned on depression datasets have shown strong performance by leveraging large-scale unlabeled speech representations.

Multimodal approaches have also gained traction. The DAIC-WOZ corpus includes audio, video, and text; many state-of-the-art systems fuse speech, linguistic, and visual cues. Text-based models using interview transcripts (sentiment, word choice, syntactic complexity) and facial action units (AUs) from video frequently outperform audio-only baselines on benchmark leaderboards. However, audio-only methods remain attractive because they are less invasive, easier to deploy, and raise fewer privacy concerns than video.

Parallel to model advances, Explainable AI (XAI) has become a priority in mental health applications. Techniques such as Grad-CAM, SHAP, LIME, and attention visualization aim to reveal which input regions or features drive predictions. This responds to a core clinical requirement: mental health professionals cannot adopt black-box systems that output labels without justification.

Emerging trends include:

- Real-time and passive monitoring via smartphone or telehealth platforms
- Longitudinal tracking of depressive episodes through repeated speech samples
- Fairness and bias auditing across gender, age, language, and accent
- Clinical validation studies involving psychiatrists and decision-support workflows
- Federated and privacy-preserving learning for sensitive health data
- Integration with digital phenotyping and wearable sensor data

Future research is likely to move from proof-of-concept classification toward trustworthy, interpretable, and clinically validated decision-support tools rather than standalone diagnostic replacements.

2.2.3 Issues and research challenges

Despite progress, several issues persist:

1. Dataset limitations — Most studies use small or single-corpus datasets. DAIC-WOZ contains 189 participants, but many published experiments use subsets. Generalization across recording conditions, interview protocols, and populations remains unproven.
2. Label reliability and subjectivity — Depression labels from PHQ-8 thresholds or clinician judgment introduce noise. Binary classification oversimplifies a spectrum disorder.
3. Evaluation methodology — Segment-level evaluation with overlapping windows inflates sample size and ignores correlation within speakers. Participant-level (speaker-independent) splits are essential but not universally applied.

4. Class imbalance and demographic bias — Depressed and non-depressed classes are often imbalanced; models may learn speaker identity or demographic artifacts rather than pathology.
5. Explainability gap — Many high-performing deep models remain opaque. Existing XAI methods often produce visual heatmaps without clinically meaningful, time-stamped explanations tied to known depressive speech markers (e.g., low energy, increased pauses).
6. Clinical adoption barriers — Regulatory requirements, ethical concerns (false positives/negatives), and lack of prospective clinical trials limit deployment.
7. Multimodal vs. unimodal trade-offs — Multimodal systems perform better but are harder to deploy; unimodal speech systems are simpler but may sacrifice accuracy.

These challenges define the context within which the present research is situated.

2.3 Comparative Analysis of Existing Methods

Existing approaches can be compared across several dimensions: feature representation, model architecture, modality, explainability, dataset, and evaluation protocol. Table 2.1 summarizes representative method categories (you may format this as a proper table in your thesis).

Approach	Features / Input	Model	Modality	Explainability	Typical Strengths	Typical Weaknesses
Handcrafted acoustic + classical ML	MFCCs, pitch, energy, pauses, COVAREP	SVM, RF, GMM	Audio	High (features are interpretable)	Transparent, low data need	Manual feature design, limited representation power
Engineered features + deep MLP	openSMILE / COVAREP feature vectors	MLP, DNN	Audio	Moderate	Strong baselines on DAIC-WOZ	Still relies on fixed feature sets
CNN on spectrograms	Mel / log-mel spectrograms	CNN	Audio	Low (black-box)	Learns time–frequency patterns automatically	Needs sufficient data; opaque decisions
RNN / LSTM on	Frame-level acoustic features	LSTM,	Audio	Low–Moderate	Models temporal dynamics	Computationally heavier; data-hungry

sequences		BiLSTM				
CNN–LSTM hybrid	Spectrogram or feature sequences	CNN + LSTM	Audio	Low	Combines local and global patterns	Complex training; hard to interpret
Pre-trained speech models	Raw waveform or representations	wav2vec, HuBERT + classifier	Audio	Low	Strong transfer learning	Large compute; explanation harder
Multimodal fusion	Audio + text + video	Various fusion networks	Multimodal	Low	Highest benchmark scores	Privacy, complexity, deployment cost
XAI-enhanced CNN	Mel spectrograms + acoustic features	CNN + Grad-CAM / SHAP	Audio	High	Classification + visual/feature explanations	Extra pipeline complexity; validation of explanations often lacking

Table 2.1

Comparative observations:

- Traditional ML offers interpretability but often underperforms deep learning on large corpora.
- CNN-on-spectrogram methods align with speech emotion recognition literature and are efficient for segment-level analysis.
- Multimodal systems dominate leaderboards but are not always suitable for lightweight, speech-only screening.
- XAI integration remains the least mature area; few systems provide second-level timeline explanations linked to clinical acoustic markers.

The proposed framework occupies the CNN + explainability quadrant: it adopts a proven spectrogram-based classifier while adding Grad-CAM, acoustic feature importance, and time-stamped natural-language explanations—addressing a gap left by purely predictive models.

2.4 Strengths and Limitations of Current Approaches

Strengths

1. Non-invasive assessment — Speech can be collected via standard microphones without specialized clinical equipment.
2. Objective quantification — Acoustic measures reduce sole reliance on self-report, which is affected by stigma and recall bias.
3. Demonstrated feasibility — Consistent moderate-to-strong results on DAIC-WOZ and related corpora confirm discriminative signal in voice.
4. Deep learning representational power — CNNs and pre-trained models capture subtle patterns not encoded in hand-crafted features.
5. Multimodal enrichment — Combining speech with text and facial cues improves robustness when full interview data is available.
6. Growing explainability toolkit — Grad-CAM, SHAP, and attention mechanisms enable post-hoc interpretation of model decisions.

Limitations

1. Black-box predictions — Most deep learning depression detectors output a label or score without clinician-usable reasoning.
2. Small-sample overfitting — Many studies, especially student/thesis prototypes, use very small participant counts, yielding unreliable generalization claims.
3. Segment-level evaluation bias — Reporting accuracy on overlapping 3-second windows overstates effective sample size.
4. Single-corpus dependency — Models trained on DAIC-WOZ may not transfer to different languages, accents, or interview styles.
5. Weak clinical validation — Few systems are evaluated in prospective settings with mental health professionals assessing utility and trust.
6. Explanation–clinical alignment — XAI heatmaps on spectrograms are not always mapped to seconds, symptoms, or known biomarkers (pitch flatness, pause ratio, energy).
7. Ethical and regulatory gaps — False positives and negatives carry real harm; most systems lack FDA/CE-style validation or clear decision-support framing.

2.5 Summary of Issues

The literature review reveals a field that has progressed from observational clinical insights to sophisticated deep learning pipelines, yet several systemic issues remain unresolved:

- Depression detection from speech is scientifically plausible but clinically unvalidated at scale.
- Performance metrics are often inflated by improper data splitting and correlated segments.

- Explainability is frequently treated as an optional add-on rather than a core design requirement.
- Deployment-ready systems that combine accurate classification, interpretable output, and a usable interface are rare.
- Ethical safeguards (informed consent, bias auditing, decision-support disclaimers) are inconsistently addressed.

These issues collectively indicate that the next meaningful contribution is not merely another classifier, but an integrated, explainable framework that clinicians and researchers can inspect, critique, and iteratively improve.

2.6 Discussion on Research Gaps

A clear gap exists between predictive performance and clinical trustworthiness. State-of-the-art models may achieve strong scores on benchmark datasets while remaining unsuitable for real-world mental health workflows because they cannot answer: Why was this person flagged? Which moments in the recording matter? Which acoustic symptoms were detected?

Furthermore, most audio-only systems either:

- (a) maximize accuracy with opaque deep networks, or
- (b) use interpretable handcrafted features with limited learning capacity.

Few works unify a deep spectrogram-based classifier with multi-level explainability—visual (Grad-CAM on mel spectrograms), feature-level (pitch, energy, pauses, MFCCs), and temporal (exact-second timeline cards)—within a deployable decision-support application.

Additional gaps include:

- Lack of participant-level cross-validation in small-scale studies
- Absence of standardized explainability metrics for mental health AI
- Limited exploration of depression subtype profiling from acoustics (with appropriate non-diagnostic framing)
- Insufficient comparison between explainable and non-explainable variants of the same backbone architecture

The present research is designed to address these gaps directly.

2.6.1 Definition of the Research Gap/Problem

Research gap: Existing speech-based depression detection systems predominantly function as black-box classifiers. Although deep learning models—particularly CNNs operating on mel spectrograms—can learn discriminative time–frequency patterns, they do not provide clinicians with transparent, time-localized, and acoustically grounded explanations for their predictions. Concurrently, interpretable systems based on handcrafted features lack the representational flexibility of deep learning. There is therefore a methodological and practical gap for an integrated framework that combines accurate binary depression classification with explainable AI techniques suitable for clinical decision support.

Problem statement: Major Depressive Disorder and related conditions affect a large global population, yet diagnosis depends heavily on subjective clinical assessment and self-report instruments that are time-consuming and variable in quality. Speech carries paralinguistic markers of depression—including reduced pitch variability, lower vocal energy, increased pause duration, and slower articulation—but automated tools that exploit this signal either lack explainability or lack rigorous, leakage-aware evaluation on clinical interview data.

Specific problem addressed by this thesis: Design, implement, and evaluate an explainable deep learning framework that:

1. Detects depression from clinical interview speech using a CNN trained on mel-spectrograms (DAIC-WOZ–style data);
2. Explains predictions via Grad-CAM heatmaps, gradient-based acoustic feature importance, and second-level timeline explanations;
3. Aggregates segment-level inference into recording-level decision support;
4. Optionally provides research-oriented depression subtype profile matching;
5. Deploys the pipeline as a web-based application for upload, prediction, and visualization.

Research questions (aligned with the gap):

- RQ1: Can a CNN trained on mel-spectrogram representations reliably classify depressive vs. non-depressive speech under participant-level data splitting?
- RQ2: Can XAI methods (Grad-CAM and feature importance) identify time regions and acoustic properties consistent with known depressive speech markers?
- RQ3: Does integrating explainability into the inference pipeline improve transparency without degrading classification performance relative to the same backbone model?

Scope boundaries (important to state explicitly):

- Binary classification (depressed vs. non-depressed); not a clinical diagnosis
- Speech-only; transcripts and video excluded from the primary model
- Small DAIC-WOZ–style subset (9 participants, 108 segments); results are preliminary
- Subtype module is heuristic profile matching, not supervised subtype classification

2.7 Summary

This chapter reviewed the evolution of speech-based depression detection from early acoustic–phonetic studies and handcrafted feature classifiers to modern deep learning and multimodal fusion approaches. Recent work demonstrates that voice contains measurable depression-related signal, especially when analyzed through time–frequency representations such as mel spectrograms. CNNs and pre-trained

speech models have improved predictive capability, while explainable AI has emerged as a necessary complement for clinical acceptance.
Comparative analysis shows a trade-off between interpretability (traditional features + classical ML) and performance (deep learning black boxes). Multimodal systems achieve strong benchmark results but increase complexity and privacy concerns. Evaluation practices—particularly speaker-independent splitting and participant-level metrics—remain inconsistent across the literature.
The central unresolved issue is the explainability gap: clinicians require understandable, time-localized justification for automated flags, but most existing systems do not provide it in a clinically meaningful form. Section 2.6 defined this as the primary research gap motivating the present thesis.
The following chapter describes the proposed Explainable Deep Learning Framework for Speech-Based Depression Detection, which integrates a three-layer CNN classifier, Grad-CAM visualization, acoustic feature analysis, timeline explanations, and a web-based deployment—positioned explicitly as a decision-support research prototype rather than a standalone diagnostic instrument.

CHAPTER 3
THEORETICAL FOUNDATIONS FOR THE EXTENSION

3.1 Introduction

Chapter 2 established that speech-based depression detection has matured from handcrafted acoustic analysis to deep learning, yet a critical limitation persists: most systems prioritize predictive accuracy over interpretability. Clinicians and researchers cannot fully trust models that output a binary label without indicating why, where, and through which acoustic mechanisms a prediction was made.
This chapter presents the theoretical foundations for the proposed extension to conventional speech-based depression detection. The extension does not replace the core classification paradigm—Convolutional Neural Networks (CNNs) operating on mel-spectrogram representations remain the primary predictive engine. Rather, it extends the baseline architecture with a multi-layer explainability framework grounded in established theories from clinical psychiatry, paralinguistics, deep learning, and explainable artificial intelligence (XAI).
The objectives of this chapter are to:

1. Summarize the theoretical bases of existing speech-based depression detection;
2. Revisit the research gap identified in Chapter 2;
3. Justify why an explainability extension is necessary and feasible;
4. Describe the theoretical principles that inspire each component of the extension;

5. Characterize the extension formally and show how it bridges the identified gap.

This chapter thus transitions from literature critique (Chapter 2) to the conceptual blueprint that underpins the system design, implementation, and evaluation presented in subsequent chapters.

3.2 Overview of Existing Theory

Speech-based depression detection rests on several interrelated theoretical domains. Understanding these is essential before introducing the extension.

3.2.1 Clinical Theory: Depression and Psychomotor Retardation

Major Depressive Disorder (MDD) is characterized by persistent low mood, anhedonia, fatigue, and cognitive slowing. A clinically relevant manifestation is psychomotor retardation— observable slowing of thought, speech, and movement. In speech, this manifests as:

- Reduced articulation rate and longer response latencies
- Increased pause duration and silence ratio
- Lower vocal energy and reduced dynamic range
- Flatter prosody (reduced pitch variability)

These symptoms are codified in diagnostic frameworks such as the DSM-5 and measured quantitatively through instruments like the PHQ-8 (Patient Health Questionnaire), which provides the ground-truth labels used in corpora such as DAIC-WOZ. The clinical theory establishes that depression is not merely lexical—it alters the production of speech at the acoustic and temporal level.

3.2.2 Paralinguistic and Acoustic–Phonetic Theory

Paralinguistics studies how something is said, independent of what is said. Key constructs include:

Construct	Acoustic Measure	Depression-Related Direction
Prosody	Pitch mean, pitch std	Lower variability (monotone)
Intensity	RMS energy, energy std	Reduced loudness and range
Temporal structure	Speech rate, pause ratio	Slower, more silent
Voice quality	Jitter, shimmer, spectral centroid	Altered timbre
Spectral envelope	MFCCs, mel bands	Captured in cepstral/spectral features

Table 3.1

The Source–Filter Model of speech production explains that the glottal source (pitch, energy) and vocal tract filter (formants, spectral shape) jointly determine perceived voice quality. Depression-related psychomotor changes affect both source (weaker, less variable excitation) and temporal organization (longer gaps between utterances).

3.2.3 Signal Processing Theory: Time–Frequency Representations

Speech is a non-stationary signal; its frequency content evolves over time. The Short-Time Fourier Transform (STFT) computes localized spectra using a sliding window. Mel-scale spectrograms apply a perceptually motivated frequency warping (mel filterbank) to emphasize bands relevant to human hearing.
Mel-spectrograms provide a two-dimensional representation (frequency × time) suitable for image-like processing. Log-power compression (decibels) and per-segment normalization stabilize dynamic range across speakers. This representation is theoretically justified because depression-related cues—pitch contours, energy envelopes, pause patterns—manifest as structured patterns across both axes.

3.2.4 Machine Learning Theory: Deep Convolutional Classification

Convolutional Neural Networks exploit translation equivariance: filters learned at one time–frequency location generalize across the spectrogram. For depression detection:

- Early conv layers detect local spectral textures and short temporal events
- Deeper layers compose hierarchical patterns (e.g., sustained low-energy regions)
- Global pooling and fully connected layers aggregate segment-level evidence into a scalar depression logit

The Universal Approximation Theorem and empirical success of CNNs in speech emotion recognition support their use as the baseline classifier. However, standard CNNs are functionally opaque—their internal representations are not directly mapped to clinical constructs.

3.2.5 Evaluation Theory: Speaker Independence and Segment Aggregation

Proper evaluation requires participant-level (speaker-independent) splitting: all segments from one individual must reside in a single partition. Without this, models may learn speaker identity rather than pathology—a form of data leakage.
At inference, segment-level probabilities are aggregated (e.g., by arithmetic mean) into a recording-level decision. This follows the theory that depression is a global state expressed locally across an interview, and that overlapping 3-second windows

with 50% hop provide a dense temporal sampling of paralinguistic behavior without losing boundary information.

3.2.6 Explainable AI Theory (Existing but Underapplied)

Post-hoc explanation methods include:

- Grad-CAM (Gradient-weighted Class Activation Mapping): uses gradients flowing into the final convolutional layer to produce a spatial importance map over the input
- Feature attribution (SHAP, integrated gradients): quantifies each input feature's contribution to the output
- Attention mechanisms: learn explicit weighting over temporal steps (more common in sequence models)

In mental health AI, trust calibration theory holds that explanations must be faithful (reflect model reasoning), understandable (accessible to non-ML experts), and actionable (linked to inspectable evidence such as timestamps or known biomarkers). Existing depression detection literature applies XAI inconsistently and rarely integrates it into deployable clinical workflows.

3.3 Revisiting the Research Gap

Chapter 2 identified a persistent gap between prediction and interpretation. Revisiting it in theoretical terms:

Gap 1 — Representational opacity:
CNNs on mel-spectrograms learn discriminative time–frequency patterns, but these patterns exist in a high-dimensional latent space with no guaranteed alignment to clinical acoustic markers (pitch flatness, pause ratio, low energy).

Gap 2 — Temporal localization deficiency:
Many systems output a single label for an entire recording. Even segment-level models rarely communicate which seconds most influenced the decision in a clinician-readable format.

Gap 3 — Dual-path disconnect:
Interpretable handcrafted-feature classifiers and opaque deep models are often treated as alternatives rather than complementary components within a unified framework.

Gap 4 — Deployment and trust gap:
Research prototypes typically stop at accuracy tables. There is no end-to-end pipeline from upload → prediction → multi-modal explanation → decision-support interface.

Gap 5 — Evaluation of explainability:
Classification metrics (Accuracy, Precision, Recall, F1) are well standardized; metrics for explanation quality (fidelity, clinical plausibility, user trust) are not.

Formal gap statement:
There exists no integrated, theoretically grounded extension to standard CNN-based speech depression classifiers that simultaneously preserves classification performance and delivers multi-level, time-stamped, clinically aligned explanations suitable for decision-support deployment.
This gap motivates the extension described in the remainder of this chapter.

3.4 Rationale for the Extension

The extension is justified on four grounds: clinical, technical, ethical, and practical.

3.4.1 Clinical Rationale

Mental health professionals operate under evidence-based decision-making. A screening tool that states "Depressed (71% confidence)" without supporting evidence fails the minimum standard for clinical adjunct systems. An extension that cites exact time ranges (e.g., 16s–19s), associated acoustic cues (low energy, long pauses), and spectrogram regions provides inspectable evidence aligned with how clinicians already listen for psychomotor signs during interviews.

3.4.2 Technical Rationale

Explainability need not sacrifice accuracy. Post-hoc XAI methods such as Grad-CAM operate on a trained model without altering its weights. A parallel FeatureClassifier (MLP on 23 acoustic features) provides a complementary attribution pathway grounded in interpretable paralinguistic theory. This dual-path design follows the principle of complementary explanations: visual (where the CNN looked) and semantic (which known features mattered).

3.4.3 Ethical and Responsible AI Rationale

Responsible AI frameworks (e.g., WHO guidance on ethics and governance of AI for health) emphasize transparency, accountability, and human oversight. For depression detection—where false positives cause anxiety and false negatives delay treatment—users and clinicians must understand system limitations. The extension embeds disclaimers and surfaces uncertainty (varying confidence scores, opposing segments) rather than presenting a single opaque verdict.

3.4.4 Practical Rationale

A web-deployable extension transforms a training script into a usable research instrument. FastAPI backend inference, timeline visualizations, and Grad-CAM plots enable psychiatrists, researchers, and thesis evaluators to interact with the system directly—supporting validation, demonstration, and iterative refinement.

3.5 Theoretical Foundations/Inspiration for Extension

Each extension component is inspired by a specific theoretical principle.

3.5.1 Baseline Classifier: CNN on Mel-Spectrograms

Foundation: Image-like deep learning on time–frequency representations (speech emotion recognition literature).
Inspiration: Treat the log-mel spectrogram as a 2D input (1, 128, T) where convolutions detect local depression-related patterns—sustained low-energy bands, reduced pitch modulation visible as horizontal striations, extended silent gaps.
Architecture theory: Three convolutional blocks (32 → 64 → 128 filters) with BatchNorm, MaxPooling, Dropout (0.4), and AdaptiveAvgPool implement hierarchical feature extraction with regularization appropriate for small datasets.

3.5.2 Grad-CAM Extension

Foundation: Selvaraju et al.'s Grad-CAM — gradients of the target class score with respect to final conv-layer feature maps produce a spatial heatmap of importance.
Extension-specific innovation: Grad-CAM is computed on the third convolutional layer (the Grad-CAM target layer) and mapped back to absolute seconds in the recording, not abstract frame indices. Peak influence time is annotated (e.g., "Peak at 16s"), bridging deep feature space and clinical temporal listening.
Theoretical requirement: Localization fidelity — the heatmap should highlight time–frequency regions that, if perturbed, would most change the prediction.

3.5.3 Acoustic Feature Pathway and Feature Importance

Foundation: Paralinguistic theory and classical acoustic depression biomarkers.
23 interpretable features extracted per segment:

- Pitch mean/std (librosa.piptrack)
- Energy mean/std (RMS)
- Speech rate, pause ratio
- Zero-crossing rate, spectral centroid/rolloff/bandwidth
- MFCC 1–13 means

A secondary FeatureClassifier (MLP: 23 → 64 → 32 → 1) is trained alongside the CNN. Gradient-based feature importance ranks features by mean absolute gradient of the output with respect to standardized inputs—an approximation inspired by integrated gradients and SHAP, chosen for inference speed.

Theoretical role: This pathway anchors explanations in clinically named constructs (monotone pitch, low energy, slow speech) rather than raw mel-band activations alone.

3.5.4 Timeline Explanation Layer

Foundation: Temporal segmentation theory and psycholinguistic event analysis.
Mechanism: Audio is segmented into 3-second windows with 50% overlap. Each segment receives an independent CNN probability and acoustic feature vector. Segments are ranked by support for the final prediction; top supporting and opposing segments generate natural-language explanations via rule-based mapping from feature thresholds to symptom tags:

Feature Condition	Generated Cue
pitch_std < 30 Hz	Flat/monotone pitch
energy_mean < 0.015	Low voice energy
pause_ratio > 0.4	Long pauses
speech_rate < 0.5	Slow speech rate

Table 3.2

Theoretical basis: Template-based semantic grounding—linking continuous acoustic measurements to discrete, clinician-recognizable descriptors derived from depression speech literature.

3.5.5 Recording-Level Aggregation

Foundation: Ensemble and evidence-combination theory.
Final prediction = mean of segment probabilities. Confidence reflects agreement across segments. Summary text cites the fraction of segments scoring above threshold (e.g., "7 of 12 segments showed depression-linked patterns") and references the strongest time ranges.
This follows the theory that depression manifests as a pervasive but locally variable acoustic signature across an interview—not a single isolated cue.

3.5.6 Depression Subtype Profile Matcher (Optional Extension)

Foundation: Clinical typology of depressive disorders (MDD, Dysthymia, Bipolar, SAD, Postpartum, Psychotic, Situational).

Mechanism: Gaussian profile matching against idealized acoustic symptom vectors—not supervised subtype learning (no subtype labels in DAIC-WOZ).
Theoretical position: Exploratory hypothesis-generation tool only; explicitly not a diagnostic subtype classifier. Grounded in the theory that different depression subtypes may exhibit distinct acoustic profiles, but evidence from speech alone is insufficient for subtype diagnosis.

3.5.7 Deployment Layer

Foundation: Human–computer interaction and clinical decision-support system (CDSS) theory.
A CDSS should augment, not replace, clinician judgment. The FastAPI + web frontend architecture implements the pipeline theory: transparent stages (preprocess → segment → classify → explain → visualize) that users can inspect at each step.

3.6 Characteristics of the Extension

The proposed extension can be formally characterized by the following properties:

Characteristic	Description
Non-invasive add-on	Explainability layers wrap the CNN without modifying its core classification function
Multi-level explanation	Model-level (Grad-CAM), feature-level (23 acoustic attributes), temporal-level (timeline cards), and summary-level (natural language)
Dual-model architecture	DepressionCNN (spectrogram path) + FeatureClassifier (acoustic path) operating in parallel
Time-localized output	All visualizations use absolute seconds in the recording
Segment-then-aggregate inference	3s windows, 50% overlap; mean probability aggregation
Clinically grounded semantics	Explanations reference known depression speech markers
Decision-support framing	Explicit non-diagnostic disclaimers embedded in outputs
Deployable end-to-end system	Training (train.py), inference (predict.py), API (server.py), and web UI
Leakage-aware evaluation	Participant-level train/test split (22% holdout)
Scalable XAI trade-off	Gradient importance by default; SHAP available but disabled for speed

Table 3.3

Architectural Summary

```
Input Audio (16 kHz mono)
       |
       ├─▶ Mel Spectrogram ─▶ DepressionCNN ─▶ P(depressed)
       |                              |
       |                              └─▶ Grad-CAM heatmap
       |
           └─▶ 23  Acoustic  Features  ─▶  FeatureClassifier  ─▶  Feature
importance
                 |
                 └─▶ Rule-based cue extraction ─▶ Timeline explanations
                         |
                         └─▶ Optional subtype profile matching
                                  |
                                  ▼
                        Web UI: prediction + visualizations
```

Differentiation from Baseline

A baseline in this context is a CNN that outputs only P(depressed) per segment/recording. The extension adds six capabilities the baseline lacks:

1. Grad-CAM spectrogram highlighting with peak-second annotation
2. Full-recording spectrogram with color-coded regions
3. Voice timeline probability chart
4. Ranked acoustic feature importance
5. Natural-language time-stamped explanations
6. Optional depression subtype profile ranking

3.7 Bridging the research gap with extension

The extension maps directly onto each gap identified in Section 3.3:

Research Gap	Extension Component	How the Gap Is Bridged
Representational opacity	Grad-CAM on final conv layer	Visualizes which time–frequency regions drove the CNN decision
Temporal localization deficiency	3s segmentation + timeline cards + peak-second Grad-CAM	Clinicians see exact seconds (e.g., 16s–19s)

		with supporting/opposing evidence
Dual-path disconnect	Parallel CNN + FeatureClassifier with shared segment pipeline	Deep learning power + interpretable acoustic semantics in one framework
Deployment and trust gap	FastAPI web app with 6 explanation tabs	End-to-end upload-to-explanation workflow for real users
Lack of clinical semantic alignment	Rule-based cue mapping (pitch, energy, pauses → symptom tags)	Explanations use clinically meaningful language, not raw tensors
Evaluation limitations	Participant-level split + segment aggregation + confidence reporting	Honest evaluation protocol; uncertainty surfaced in UI

Table 3.4

Theoretical Closure

The extension achieves theoretical closure by unifying three previously separate research threads:

1. Clinical psychomotor theory → defines what should be explained (pitch, energy, pauses)
2. Deep learning classification theory → provides accurate segment-level detection
3. Post-hoc XAI theory → provides mechanisms to expose model reasoning

Without any one thread, the system would be incomplete: clinical theory without ML lacks automation; ML without XAI lacks trust; XAI without clinical grounding lacks meaning.

Alignment with Research Questions

Research Question	Theoretical Bridge
RQ1: Can deep learning detect depression from speech?	CNN + mel-spectrogram theory; evaluated with participant-level F1
RQ2: Does XAI improve transparency?	Grad-CAM + feature importance + timeline theory
RQ3: Does explainability degrade performance?	Extension is post-hoc; same CNN backbone preserved

Table 3.5

3.8 Summary

This chapter established the theoretical foundations for extending conventional CNN-based speech depression detection with a comprehensive explainability framework. Existing theory spans clinical psychomotor retardation, paralinguistic acoustics, time–frequency signal processing, deep convolutional learning, speaker-independent evaluation, and post-hoc XAI—each necessary but insufficient alone.

The research gap was revisited as a disconnect between accurate prediction and clinically meaningful interpretation. The extension was justified on clinical, technical, ethical, and practical grounds. Its components—Grad-CAM, acoustic feature attribution, timeline semantic grounding, recording-level aggregation, optional subtype profiling, and web deployment—each derive from explicit theoretical principles.

The extension is characterized as a non-invasive, multi-level, dual-path, time-localized, decision-support-oriented augmentation to a standard CNN baseline. Section 3.7 demonstrated how each identified gap is bridged by a specific extension component, achieving theoretical closure across clinical, machine learning, and explainability domains.

Chapter 4 will translate these theoretical foundations into the system design and methodology—detailing the preprocessing pipeline, model implementation, training protocol, XAI integration, and evaluation framework used in this thesis project on DAIC-WOZ–style clinical interview data (9 participants, 108 segments, binary depressed vs. non-depressed classification).

CHAPTER 4
APPROACH

4.1 Introduction

Chapters 2 and 3 established the literature context, research gap, and theoretical foundations for extending conventional speech-based depression detection with explainable artificial intelligence (XAI). This chapter describes the practical approach taken to design, implement, and deploy that extension.

The approach follows a structured pipeline: define testable hypotheses, specify inputs and outputs, model the end-to-end workflow, select technologies, characterize the extension's features, identify target users and use scenarios, and position the work within the broader AI knowledge base.

The system developed in this thesis is an Explainable Deep Learning Framework for Speech-Based Depression Detection. It accepts clinical interview audio, produces a binary depression prediction with confidence, and returns multi-level explanations—Grad-CAM heatmaps, acoustic feature importance, time-stamped timeline cards, and optional depression subtype profile rankings—through a web-based decision-support interface.

This chapter focuses on how the extension is approached and organized. Detailed model architecture, training hyperparameters, and evaluation results are covered in subsequent chapters.

4.2 Hypothesis and its inspiration

4.2.1 Primary Hypothesis

H_1: A Convolutional Neural Network (CNN) trained on log-mel spectrograms can reliably distinguish depressed from non-depressed speech in clinical interview

recordings when evaluated under participant-level (speaker-independent) data splitting.
Inspiration: Clinical and paralinguistic research consistently links depression to measurable acoustic changes—reduced pitch variability, lower vocal energy, increased pause duration, and slower articulation (psychomotor retardation). Deep CNNs have demonstrated success in learning time–frequency patterns from mel-spectrogram representations in speech emotion and affect recognition. The DAIC-WOZ corpus validates depression labels via PHQ-8 clinical assessments, providing a grounded training signal.

4.2.2 Explainability Hypothesis

H_2: Post-hoc explainability techniques—Grad-CAM on the CNN and gradient-based feature importance on an acoustic FeatureClassifier—can identify time regions and speech characteristics that align with clinically recognized depressive markers, thereby improving system transparency without altering classification performance.
Inspiration: The trust gap identified in Chapter 2: clinicians reject black-box predictions. Grad-CAM (Selvaraju et al.) provides spatial importance maps over spectrogram inputs. Parallel interpretable acoustic features (pitch, energy, pauses, MFCCs) bridge deep representations and clinical vocabulary. The dual-path design is inspired by complementary explanation theory—visual where (CNN) plus semantic what (acoustic features).

4.2.3 Temporal Localization Hypothesis

H_3: Segmenting recordings into overlapping 3-second windows and aggregating segment-level probabilities enables time-localized explanations that cite exact seconds in the voice recording, making model output inspectable at the moment level.
Inspiration: Depression manifests locally within an interview—certain responses may show stronger psychomotor slowing than others. Fixed-length segmentation with 50% overlap is standard in speech affect research and ensures boundary coverage. Mean-probability aggregation treats depression as a pervasive state expressed across multiple local acoustic events.

4.2.4 Deployment Hypothesis

H_4: Packaging the classification and explanation pipeline as a web application (FastAPI backend + interactive frontend) makes the extension usable as a research and decision-support prototype by non-technical stakeholders.
Inspiration: Clinical Decision Support System (CDSS) theory requires systems to augment—not replace—clinician judgment. A deployable interface allows direct inspection of predictions and explanations, supporting demonstration, feedback, and future clinical validation studies.

4.2.5 Null Expectations (Explicit Limitations)

The approach explicitly does not hypothesize clinical-grade diagnostic accuracy. With nine DAIC-WOZ–style participants (108 segments), results are treated as proof-of-concept for the pipeline, not population-level generalization. Subtype profile matching is hypothesized only as a research exploration tool, not as validated subtype diagnosis.

4.3 Inputs and Outputs of Extension

4.3.1 Inputs

Input Category	Specification	Purpose
Primary input	Voice recording (WAV preferred; also MP3, FLAC, OGG, M4A, WebM)	Source speech signal
Audio properties	Resampled to 16 kHz, mono	Standardized processing
Training data	DAIC-WOZ–style interviews: 9 participants (5 depressed, 4 non-depressed)	Supervised learning
Labels	Binary: Depressed (1) vs Non-Depressed (0), PHQ-8–validated	Ground truth
Optional context	User checkboxes: chronic mood, recent stress, postpartum, seasonal, mood swings	Subtype profile boosting (optional)

Table 4.1

Per-segment internal inputs (after preprocessing):

- Log-mel spectrogram: shape (1, 128, T) — CNN input
- Acoustic feature vector: 23 dimensions — FeatureClassifier input

23 acoustic features:
pitch_mean_hz, pitch_std_hz, energy_mean, energy_std, speech_rate, pause_ratio, zero_crossing_rate, spectral_centroid, spectral_rolloff, spectral_bandwidth, mfcc_1_mean through mfcc_13_mean

4.3.2 Outputs

Output	Format	Description
Prediction	Categorical	"Depressed" or "Non-Depressed"
Confidence	Float (0–1)	Distance from decision threshold (0.5)
Probability (depressed)	Float (0–1)	Mean segment-level CNN probability
Segment probabilities	List of floats	Per 3-second window depression score
Timeline explanations	List of structured cards	Exact time range, probability, role (supporting/opposing), natural-language text, symptom cues

Prediction reason	Natural language	Summary citing segment counts and key time ranges
Grad-CAM heatmap	2D array + visualization	Time–frequency importance for key segment
Full spectrogram	2D array + time axis	Entire recording with highlighted regions
Highlight regions	List of {start_sec, end_sec, prob, role}	Segments with strong depression/non-depression signal
Feature importance	Dict (feature → score)	Ranked acoustic contributors
Acoustic features	Dict (23 values)	Raw features for key segment
Subtype ranking	List of 7 profiles with probabilities	Optional research output
Explanation summary	Markdown text	Combined report with disclaimer

Table 4.2

Saved model artifacts (training output):

File	Contents
depression_cnn.pt	CNN weights (~1.4 MB)
feature_model.pt	MLP weights (~18 KB)
feature_scaler.pkl	StandardScaler for 23 features
training_metadata.json	Segment count, participant count, test metrics

Table 4.3

4.4 Process workflow for Extension

The extension follows two workflows: offline training and online inference.

4.4.1 Training Workflow

STEP 1: DATA DISCOVERY
Scan depressed/ and non depressed/ folders
Locate *_AUDIO.wav per participant (9 files total)

▼

STEP 2: PREPROCESSING
Load at 16 kHz mono → trim silence (top_db=25)
Peak normalize → cap at 60 seconds per file
Segment into 3s windows (50% overlap, max 12 segments)
Result: 108 segments (60 depressed, 48 non-depressed)

▼

STEP 3: FEATURE EXTRACTION
Per segment: log-mel spectrogram (128 bands, z-score norm)
Per segment: 23 acoustic features

▼

STEP 4: PARTICIPANT-LEVEL SPLIT
22% of participants held out for testing (~2 participants)
All segments from one person stay in same split

▼

STEP 5: CNN TRAINING
DepressionCNN | Adam lr=1e-3 | BCEWithLogitsLoss
12 epochs, batch size 8, weight decay 1e-4
Monitor F1 on test set every 4 epochs

▼

STEP 6: EVALUATION
Accuracy, Precision, Recall, F1 on held-out test segments

▼

STEP 7: DEPLOYMENT RETRAINING
Load best weights → 8 additional epochs on full dataset

```
│ Save depression_cnn.pt                    │
└──────────────────────────────────────────────────────┬
──────────────────────────┘
                    ▼
┌──────────────────────────────────────────────────────
──────────────────────────┐
│ STEP 8: FEATURE MODEL TRAINING                          │
│ StandardScaler on 23 features → FeatureClassifier MLP   │
│ 30 epochs, Adam lr=1e-3                      │
│ Save feature_model.pt + feature_scaler.pkl          │
└──────────────────────────────────────────────────────┬
──────────────────────────┘
                    ▼
┌──────────────────────────────────────────────────────
──────────────────────────┐
│ STEP 9: METADATA EXPORT                             │
│ Save training_metadata.json with test metrics       │
└──────────────────────────────────────────────────────
──────────────────────────┘
```

Command: python3 train.py

4.4.2 Inference Workflow (Extension Pipeline)

User uploads audio via Web UI

```
      │
      ▼
┌──────────────────────────────────────────────────────
──────────────────────────┐
│ 1. AUDIO INGESTION (FastAPI /api/predict)              │
│ Validate format → save to temp file → optional context flags │
└──────────────────────────────────────────────────────┬
──────────────────────────┘
                    ▼
┌──────────────────────────────────────────────────────
──────────────────────────┐
│ 2. PREPROCESSING (DepressionPredictor.predict)          │
│ load_audio: 16 kHz, trim silence, normalize         │
│ segment_audio_with_times: 3s windows, 50% overlap       │
│ compute_full_mel_spectrogram for full recording view    │
└──────────────────────────────────────────────────────┬
──────────────────────────┘
                    ▼
┌──────────────────────────────────────────────────────
──────────────────────────┐
│ 3. PER-SEGMENT CLASSIFICATION                          │
│ For each segment:                             │
```

```
│  mel spec → DepressionCNN → sigmoid → P(depressed)  │
│  audio → extract_acoustic_features → 23-dim vector  │
└─────────────────────────┬─────────────────────────┘
                          ▼
┌─────────────────────────────────────────────────────┐
│ 4. RECORDING-LEVEL AGGREGATION                      │
│ avg_prob = mean(segment probabilities)              │
│ prediction = Depressed if avg_prob ≥ 0.5 else Non-Depressed │
│ confidence = distance from 0.5                      │
└─────────────────────────┬─────────────────────────┘
                          ▼
┌─────────────────────────────────────────────────────┐
│ 5. EXPLAINABILITY LAYER                             │
│ a) build_timeline_explanations → time-stamped NL cards │
│ b) build_prediction_reason → summary paragraph      │
│ c) compute_grad_cam on key supporting segment       │
│ d) compute_feature_importance on all segment features │
│ e) highlight_regions on full spectrogram            │
│ f) classify_subtype (optional, if depressed)        │
└─────────────────────────┬─────────────────────────┘
                          ▼
┌─────────────────────────────────────────────────────┐
│ 6. RESPONSE FORMATTING & VISUALIZATION              │
│ result_to_json → frontend renders 6 tabs:           │
│  Why This Result | Depression Type | Spectrogram    │
│  Voice Timeline | Grad-CAM | Features               │
└─────────────────────────────────────────────────────┘
```

Command: python3 -m uvicorn server:app --host 127.0.0.1 --port 8765

4.4.3 Key Design Decisions in the Workflow

Decision	Choice	Rationale
Segment length	3 seconds	Captures local pitch/energy/pause patterns; standard in literature
Overlap	50%	Avoids boundary information loss

Aggregation	Mean probability	Simple, interpretable recording-level score
Grad-CAM target	3rd conv layer (128 filters)	Last spatial layer before global pool; standard Grad-CAM practice
Key segment selection	Highest-impact supporting segment	Shows strongest evidence for final prediction
Feature importance	Gradient-based (fast mode)	Real-time inference; SHAP available but disabled for speed
Decision threshold	0.5	Standard binary classification cutoff

Table 4.4

4.5 Technologies identified

The following technologies were selected to implement the extension, grouped by function.

4.5.1 Core Language and Deep Learning

Technology	Version	Role
Python	3.9+	Primary development language
PyTorch	≥ 2.0	CNN and MLP training/inference
torchvision	≥ 0.15	Supporting utilities

Table 4.5

4.5.2 Audio Processing and Feature Extraction

Technology	Version	Role
Librosa	≥ 0.10	Audio loading, mel spectrograms, pitch (piptrack), MFCCs
SoundFile	≥ 0.12	WAV I/O backend
NumPy	≥ 1.24	Numerical arrays for spectrograms and features

Table 4.6

4.5.3 Machine Learning Utilities

Technology	Version	Role

scikit-learn	≥ 1.3	StandardScaler, evaluation metrics (Accuracy, P, R, F1)
Pandas	≥ 2.0	Data handling (where applicable)
SHAP	≥ 0.43	Optional feature attribution (KernelExplainer)

Table 4.7

4.5.4 Visualization

Technology	Version	Role
Matplotlib	≥ 3.7	Spectrograms, Grad-CAM plots, timeline charts
Pillow	≥ 10.0	Image handling for API responses

Table 4.8

4.5.5 Deployment and Interface

Technology	Version	Role
FastAPI	≥ 0.104	REST API backend
Uvicorn	≥ 0.24	ASGI server
python-multipart	≥ 0.0.6	File upload handling
HTML/CSS/JavaScript	—	Web frontend (frontend/)
Streamlit	≥ 1.28	Alternative prototype UI (app.py)

Table 4.9

4.5.6 Dataset

Resource	Description
DAIC-WOZ (subset)	Clinical interview recordings with PHQ-8 labels
9 participants	5 depressed + 4 non-depressed
108 segments	3-second overlapping windows

Table 4.10

4.5.7 Technology Selection Rationale

- PyTorch over TensorFlow: dynamic computation graphs simplify Grad-CAM hook implementation on intermediate conv layers.
- Librosa over raw scipy: unified API for mel spectrograms, pitch tracking, and MFCC extraction.

- FastAPI over Flask/Django: async file upload, automatic OpenAPI docs, lightweight deployment.
- CNN over LSTM/Transformer: best bias–variance tradeoff for 9-participant dataset; Transformers require substantially more data.
- Gradient importance over SHAP (default): SHAP KernelExplainer is $O(n^2)$ and unsuitable for real-time web inference.

4.6 Features of technological extension

The extension introduces the following distinguishing features beyond a standard CNN depression classifier.

4.6.1 Multi-Level Explainability

Level	Feature	Implementation
Visual	Grad-CAM heatmap on mel spectrogram	compute_grad_cam() in explain.py
Temporal	Timeline cards with exact seconds	build_timeline_explanations()
Semantic	Natural-language cue descriptions	Rule mapping: pitch/energy/pause → symptom tags
Feature	Ranked acoustic importance	Gradient-based on FeatureClassifier
Summary	Plain-language prediction reason	build_prediction_reason()

Table 4.11

4.6.2 Dual-Model Architecture

- DepressionCNN — primary classifier on mel spectrograms (predictive path)
- FeatureClassifier — secondary MLP on 23 acoustic features (interpretation path)

Both models share the same segmented audio but serve complementary roles: accuracy and interpretability.

4.6.3 Time-Localized Evidence

Unlike systems that output a single label for an entire recording, this extension:

- Scores every 3-second window independently
- Ranks supporting and opposing segments
- Annotates Grad-CAM peak time in absolute seconds (e.g., "Peak at 16s")
- Colors full-recording spectrogram by depression signal strength

4.6.4 Interactive Web Deployment

Six frontend tabs expose the full explanation stack:

1. Why This Result? — prediction reason + timeline cards
2. Depression Type — 7 subtype profile rankings (research only)

3. Spectrogram — full recording with highlighted regions
4. Voice Timeline — probability bar chart over time
5. Grad-CAM — heatmap at key segment
6. Features — acoustic feature importance chart

4.6.5 Leakage-Aware Training Protocol

Participant-level splitting (participant_level_split(), 22% test ratio, seed 42) ensures no speaker appears in both train and test sets—a methodological feature often missing in comparable prototypes.

4.6.6 Decision-Support Safeguards

- Confidence scores reflect distance from threshold, not false certainty
- Opposing segments surfaced alongside supporting evidence
- Disclaimer embedded in every summary: "Decision-support tool only — not a clinical diagnosis"
- Subtype module explicitly labeled as research profile matching

4.6.7 Extensibility

Extension Point	Future Use
SHAP mode (fast=False)	Richer feature attribution
Context checkboxes	Subtype profile boosting
augment=True in training	Data augmentation for larger experiments
CUDA support	GPU-accelerated training/inference
Full DAIC-WOZ (189 participants)	Scale-up without architecture change

Table 4.12

4.7 Target Users / Use Scenarios

The system is designed as a research and decision-support prototype, not a consumer diagnostic product.

4.7.1 Target Users

User Group	Role	How They Use the Extension
Mental health researchers	Study speech biomarkers of depression	Upload interview recordings; inspect timeline and feature explanations
Clinical psychologists / psychiatrists	Evaluate AI-assisted screening tools	Review Grad-CAM and acoustic cues; assess whether explanations align with clinical observation
AI / ML students and academics	Learn explainable health AI pipelines	Study dual-model architecture, Grad-CAM

		integration, deployment pattern
Thesis evaluators / examiners	Assess research contribution	Run web demo; verify prediction + explanation outputs on sample audio
Telehealth platform developers	Prototype voice-based screening modules	Use FastAPI endpoints as integration reference

Table 4.13

4.7.2 Use Scenarios

Scenario 1 — Research demonstration

A researcher uploads a DAIC-WOZ interview clip. The system returns "Depressed (71% confidence)" with timeline cards showing low energy and long pauses at 22s–25s and 34s–37s. The researcher compares explanations against the clinical transcript.

Scenario 2 — Clinician inspection

A psychiatrist uploads a patient's voice sample (with consent). They examine the Grad-CAM heatmap and feature importance chart. If explanations align with their clinical impression, they gain confidence in using the tool as a supplementary signal—not a diagnosis.

Scenario 3 — Methodology teaching

An instructor demonstrates the difference between a black-box CNN (prediction only) and the extended system (prediction + 5 explanation layers) in an AI ethics or digital health course.

Scenario 4 — Pipeline validation before scale-up

Before training on the full 189-participant DAIC-WOZ corpus, the team validates preprocessing, segmentation, Grad-CAM rendering, and API response format on the 9-participant subset.

Scenario 5 — Subtype exploration (research only)

A researcher checks optional context flags (e.g., seasonal, chronic) and reviews the 7-profile subtype ranking to generate hypotheses for future labeled subtype studies.

4.7.3 Explicit Non-Use Cases

The extension must not be used for:

- Standalone clinical diagnosis without professional assessment
- Emergency mental health triage
- Employment, insurance, or legal decision-making
- Analysis of non-consenting individuals

4.8 Positioning the Extension within the AI body of knowledge

4.8.1 AI Sub-Domains

This work sits at the intersection of four AI sub-domains:

```
                    │  Health Informatics  │
                    │  (Mental Health AI)  │
                    └──────────┬───────────┘
                               │
          ┌────────────────────┼────────────────────┐
          │                    │                    │
          ▼                    ▼                    ▼
┌────────────────┐  ┌───────────────────┐  ┌──────────────────┐
│ Speech Signal  │  │ Deep Learning     │  │ Explainable AI   │
│ Processing     │  │ (CNN Classifier)  │  │ (Grad-CAM,       │
│ (Mel spec,     │  │                   │  │  Feature Attr.)  │
│  MFCC, pitch)  │  │                   │  │                  │
└────────────────┘  └───────────────────┘  └──────────────────┘
                               │
                               ▼
                    ┌──────────────────────┐
                    │  AI Deployment /     │
                    │  MLOps (FastAPI,     │
                    │  Web UI, CDSS)       │
                    └──────────────────────┘
```

4.8.2 Relationship to Existing AI Paradigms

Paradigm	This Project's Position
Supervised learning	Binary classification with PHQ-8–derived labels
Representation learning	CNN learns mel-spectrogram representations automatically
Post-hoc explainability	Grad-CAM and gradient importance applied after training
Multi-model ensemble (loose)	CNN + FeatureClassifier operate in parallel, not fused
Clinical Decision Support Systems	Web UI framed as augmentation, not autonomous diagnosis
Responsible / Trustworthy AI	Transparency, disclaimers, opposing evidence surfaced

Table 4.14

4.8.3 Contribution to AI Knowledge

Relative to the existing body of knowledge, this extension contributes:

1. Integrated XAI pipeline for mental health speech — not a standalone Grad-CAM demo, but a full upload-to-explanation workflow

2. Dual-path explanation design — CNN visual + acoustic semantic explanations in one system
3. Second-level temporal grounding — explanations cite exact seconds, not abstract frame indices
4. Deployable research prototype — bridges the gap between Jupyter notebook experiments and usable interfaces
5. Methodological rigor at small scale — participant-level splitting and explicit limitation reporting as a model for thesis-scale health AI projects

4.8.4 Comparison with Related AI Systems

System Type	Typical Output	This Extension
Black-box CNN classifier	Label + probability	Label + probability + 5 explanation layers
openSMILE + SVM baseline	Label + feature vector	Label + deep features + ranked importance
Multimodal DAIC-WOZ SOTA	Label (audio+text+video)	Label + explanations (audio-only, lighter deployment)
Commercial mental health chatbots	Text-based screening	Acoustic paralinguistic analysis with visual evidence
General-purpose XAI tools (Captum, SHAP)	Generic attributions	Domain-specific timeline cards with clinical cue mapping

Table 4.15

4.8.5 Alignment with AI Ethics Frameworks

Principle	How the Extension Addresses It
Transparency	Grad-CAM, feature charts, timeline cards
Accountability	Human clinician remains decision-maker
Fairness awareness	Limitation disclosure; small, non-diverse training set acknowledged
Privacy	Local deployment; temp files deleted after inference
Non-maleficence	Decision-support disclaimer; opposing evidence shown

Table 4.16

4.9 Summary

This chapter presented the complete approach for implementing the explainable speech-based depression detection extension. Four testable hypotheses were defined, inspired by clinical psychomotor theory, paralinguistic acoustics, XAI literature, and CDSS design principles.

The inputs (voice recordings, optional context flags) and outputs (prediction, confidence, multi-level explanations, visualizations) were specified formally. Two process workflows—offline training (train.py, 9 steps) and online inference (DepressionPredictor + FastAPI, 6 stages)—were documented with design decision rationale.

Technologies were identified across deep learning (PyTorch), audio processing (Librosa), ML utilities (scikit-learn), explainability (Grad-CAM, SHAP), and deployment (FastAPI, web frontend). Seven distinguishing features of the technological extension were characterized, including multi-level explainability, dual-model architecture, time-localized evidence, and leakag-aware evaluation.

Target users (researchers, clinicians, educators) and five concrete use scenarios were defined, alongside explicit non-use cases. The extension was positioned within the AI body of knowledge at the intersection of speech signal processing, deep learning, explainable AI, and clinical decision support—contributing an integrated, deployable, temporally grounded explanation pipeline not found in standard black-box depression classifiers.

Chapter 5 will detail the implementation—module structure, class design, API specification, and UI components—corresponding directly to the workflow and technologies described here.e

CHAPTER 5
ANALYSIS AND DESIGN

5.1 Introduction

Chapter 4 described the overall approach—hypotheses, inputs/outputs, workflows, technologies, and target users—for the explainable speech-based depression detection extension. This chapter translates that approach into a formal analysis and design specification: architectural rationale, module decomposition, data flows, interaction design, and integration within the broader AI framework.

The design follows a layered, modular architecture with three principal components:

1. Preprocessing Module — audio ingestion, normalization, and segmentation
2. ML Engine — CNN classification and FeatureClassifier for acoustic attribution
3. Extended Module — explainability, subtype profiling, API serialization, and web presentation

Separation of concerns allows the ML Engine to function as a standalone classifier while the Extended Module adds transparency without modifying core model weights. The design prioritizes inspectability, reproducibility, and deployment readiness within the constraints of a thesis-scale dataset (9 participants, 108 segments).

5.2 Rationale for the design Extension

The design extension is motivated by five requirements derived from Chapters 2–4.

5.2.1 Requirement for Modularity

A monolithic script mixing training, inference, explanation, and UI would be difficult to test, extend, or evaluate independently. The design therefore decomposes the system into discrete Python modules under src/, each with a single responsibility:

Module	Responsibility

config.py	Hyperparameters, paths, feature names
features.py	Audio loading, segmentation, feature extraction
data.py	Dataset discovery, splitting, PyTorch Dataset
model.py	CNN and MLP architectures
explain.py	Grad-CAM, timeline, visualizations
subtype.py	Depression subtype profile matching
predict.py	End-to-end inference orchestration
api_utils.py	JSON/chart serialization for the web API

Table 5.1

5.2.2 Requirement for Dual-Path Processing

Clinical trust requires both learned representations (CNN on mel spectrograms) and named acoustic biomarkers (pitch, energy, pauses, MFCCs). A single-path design forces a trade-off between accuracy and interpretability. The dual-path design runs both pipelines on the same segmented audio, merging outputs at the inference orchestration layer (DepressionPredictor).

5.2.3 Requirement for Temporal Granularity

Recording-level labels alone are insufficient for clinician inspection. The design segments audio into 3-second windows with 50% overlap, assigns independent probabilities per window, and propagates timestamps (start_sec, end_sec) through every downstream structure—timeline cards, Grad-CAM annotations, spectrogram highlights, and bar charts.

5.2.4 Requirement for Non-Invasive Explainability

Post-hoc explainability must not degrade classification performance. Grad-CAM hooks attach to the third convolutional layer during inference only; the FeatureClassifier trains separately. Neither mechanism alters the CNN's learned weights at explanation time.

5.2.5 Requirement for Deployable Decision Support

Explanations must reach end users, not remain in notebook outputs. The design introduces a client–server boundary: FastAPI exposes POST /api/predict; the frontend consumes structured JSON with embedded Base64 chart images. This follows standard MLOps separation between model logic and presentation.

5.3 Top-Level Architecture

The system architecture comprises four horizontal layers and three vertical modules.

```
┌─────────────────────────────────────────────────────────────────────────┐
│                          PRESENTATION   LAYER                          │
│   frontend/index.html   +   static/css/style.css   +   static/js/app.js │
│     (Upload  UI  ·  6  explanation  tabs  ·  result  badges  ·  charts) │
└───────────────────────────────────┬─────────────────────────────────────┘
                                    │ HTTP (REST)
┌───────────────────────────────────▼─────────────────────────────────────┐
│                     API  /  ORCHESTRATION  LAYER                        │
│        server.py   (FastAPI)     →      DepressionPredictor      →
api_utils.result_to_json│
└───────────────────────────────────┬─────────────────────────────────────┘
                                    │
             ┌──────────────────────┼──────────────────────┐
             │                      │                      │
             ▼                      ▼                      ▼
┌─────────────────┐   ┌─────────────────┐   ┌─────────────────────────┐
│  PREPROCESSING  │   │   ML ENGINE     │   │   EXTENDED MODULE       │
│  MODULE         │   │                 │   │                         │
│  features.py    │   │  model.py       │   │  explain.py             │
│  data.py        │   │  train.py       │   │  subtype.py             │
│                 │   │  models/*.pt    │   │  api_utils.py           │
└─────────────────┘   └─────────────────┘   └─────────────────────────┘
             │                      │                      │
             └──────────────────────┼──────────────────────┘
                                    │
┌───────────────────────────────────▼─────────────────────────────────────┐
```

```
│                                                              DATA   LAYER
│
│     depressed/*_AUDIO.wav    ·    non  depressed/*_AUDIO.wav    ·    models/
│
└──────────────────────────────────────────────────────────────────────────
┘
```

Architectural Principles

Principle	Application
Separation of concerns	Preprocessing, ML, and XAI are isolated modules
Single orchestrator	DepressionPredictor.predict() coordinates all inference steps
Lazy model loading	get_predictor() in server.py loads models on first request
Stateless API	Each prediction is independent; temp files deleted after response
Configuration centralization	All constants in config.py

Table 5.2

5.3.1 Preprocessing Module

The Preprocessing Module standardizes raw audio into ML-ready tensors and interpretable feature vectors. It is implemented primarily in features.py and data.py.

5.3.2 ML Engine

5.3.1.1 Components

Component	Function	Key Parameters
Audio Loader	load_audio()	16 kHz, mono, librosa.effects.trim(top_db=25), peak normalization
Segmenter	segment_audio() / segment_audio_with_times()	3.0 s duration, 50% overlap (1.5 s hop)
Mel Spectrogram Generator	compute_mel_spectrogram()	128 mel bands, FFT 2048, hop 512, log-power dB, z-score per segment

Full Mel Generator	compute_full_mel_spectrogram()	Entire recording for visualization (no per-segment normalization)
Acoustic Feature Extractor	extract_acoustic_features()	23 paralinguistic features per segment
Feature Vectorizer	features_to_vector()	Ordered float array matching FEATURE_NAMES
Dataset Builder	build_segment_dataset()	Discovers files, segments, extracts features, attaches metadata
Participant Splitter	participant_level_split()	22% holdout by participant ID, seed 42

Table 5.3

5.3.1.2 Preprocessing Pipeline Diagram

```
Raw WAV (*_AUDIO.wav)
       |
       ▼
┌────────────────────┐
│ librosa.load()     │  sr=16000, mono=True
│ trim silence       │  top_db=25
│ peak normalize     │  max(abs(y))=1.0
└─────────┬──────────┘
          |
          ├──────────────────────────────────┐
          |                                  |
          ▼                                  ▼
┌──────────────────────┐         ┌───────────────────────┐
│ Per-segment path     │         │ Full-recording path    │
│ 3s windows, 50% ovl  │         │ compute_full_mel()     │
└─────────┬────────────┘         └───────────┬───────────┘
          |                                  |
          ├──────────────┬───────────────┐   |
          ▼              ▼               ▼   ▼
   mel spec (1,128,T)  23 features   timestamps   full_mel + times
```

5.3.1.3 Design Decisions

- 16 kHz sampling — standard for speech ML; reduces compute vs. 44.1 kHz without losing paralinguistic information.
- Silence trimming — removes leading/trailing dead air that would inflate pause-ratio features.
- Per-segment z-score normalization — compensates for inter-speaker loudness differences on mel spectrograms.
- Max 12 segments / 60 s (training) — limits compute on the small dataset while preserving representative coverage via linspace subsampling.
- Max 180 s (inference) — allows longer uploads in the web app without training-time constraints.

5.3.1.4 Output Data Structures

Segment metadata (inference):

```
{
   "audio": np.ndarray,       # 3s waveform
   "start_sec": float,        # e.g., 16.0
   "end_sec": float,          # e.g., 19.0
}
```

Segment details (post-classification):

```
{
   "start_sec": float,
   "end_sec": float,
   "prob": float,             # CNN depression probability
   "features": dict,          # 23 acoustic features
}
```

5.3.2 ML Engine

The ML Engine performs supervised binary classification and auxiliary feature modeling. It is implemented in model.py, trained via train.py, and persisted under models/.

5.3.2.1 Primary Model: DepressionCNN

Purpose: Learn discriminative time–frequency patterns from mel spectrograms.
Architecture:

```
Input: (batch, 1, 128, T)
Conv2d(1→32, 3×3) + BatchNorm + ReLU + MaxPool(2×2)
Conv2d(32→64, 3×3) + BatchNorm + ReLU + MaxPool(2×2)
Conv2d(64→128, 3×3) + BatchNorm + ReLU + MaxPool(2×2)  ← Grad-CAM layer
AdaptiveAvgPool(4×4)
Flatten → 2048
```

```
Linear(2048→128) + ReLU + Dropout(0.4)
Linear(128→1) → logit
Output: sigmoid(logit) = P(Depressed)
```

Design rationale:

Layer	Purpose
3 conv blocks (32→64→128)	Hierarchical time–frequency feature learning
BatchNorm	Stabilizes training on small dataset
MaxPool	Spatial downsampling; translation tolerance
Dropout 0.4	Regularization against overfitting
AdaptiveAvgPool(4×4)	Fixed-size representation regardless of input length
BCEWithLogitsLoss	Numerically stable binary classification

Table 5.4

Grad-CAM hooks: The third convolutional layer stores activations (_grad_cam_activations) and registers a backward hook (_save_gradients) when requires_grad=True during CAM computation.

5.3.2.2 Secondary Model: FeatureClassifier

Purpose: Provide gradient-based feature importance on interpretable acoustic inputs.
Input: 23 standardized features
Linear(23→64) + ReLU + Dropout(0.3)
Linear(64→32) + ReLU
Linear(32→1) → logit
Role in design: Not the primary decision model. Trained on the same labels as the CNN but operates in the semantic feature space clinicians understand.

5.3.2.3 Training Subsystem

Phase	Description
Data load	build_segment_dataset() → 108 segments
Split	participant_level_split(test_ratio=0.22)
CNN train	12 epochs, Adam lr=1e-3, batch=8, weight_decay=1e-4
Evaluate	Accuracy, Precision, Recall, F1 on held-out participants
Deploy fine-tune	8 epochs on full dataset; save depression_cnn.pt
Feature train	StandardScaler + 30 epochs; save feature_model.pt, feature_scaler.pkl
Metadata	training_metadata.json

Table 5.5

5.3.2.4 Inference Aggregation Logic

```
For each segment i:
   prob_i = sigmoid(CNN(mel_spec_i))
avg_prob = mean(prob_1 ... prob_n)
prediction = "Depressed" if avg_prob >= 0.5 else "Non-Depressed"
confidence = avg_prob if depressed else (1 - avg_prob)
```

Threshold: 0.5 (configurable in future versions).

5.3.2.5 Model Artifact Schema

```
depression_cnn.pt:
{"cnn_state_dict": OrderedDict, "n_mels": 128}

feature_model.pt:
{"feature_model_state_dict": OrderedDict, "n_features": 23}
```

5.3.3 Extended module

The Extended Module encompasses all functionality beyond bare classification: explainability, subtype profiling, visualization, and API response formatting.

5.3.3.1 Explainability Subsystem (explain.py)

Function	Input	Output	Role
compute_grad_cam()	CNN + spectrogram	2D heatmap (128 × T)	Visual importance map
build_timeline_explanations()	segment_details, prediction	List of timeline cards	Time-stamped NL explanations
explain_voice_at_time()	features, prob, time range	String	Rule-based cue generation
build_prediction_reason()	prediction, timeline, segments	String	Summary paragraph
build_explanation_summary()	all explanation artifacts	Markdown string	Combined report
plot_full_spectrogram()	full_mel, highlight_regions	matplotlib Figure	Recording-level visualization
plot_grad_cam()	spec, cam, start_sec	matplotlib Figure	CAM with peak-second annotation
plot_timeline_chart()	segment_details	matplotlib Figure	Probability vs. time bar chart

compute_feature_importance()	FeatureClassifier, features	Dict[str, float]	Ranked acoustic contributors

Table 5.6

Timeline card structure:

```
{
   "start_sec": 16.0,
   "end_sec": 19.0,
   "time_label": "16s – 19s",
   "probability": 0.71,
   "role": "supporting" | "opposing",
   "text": "At 16s – 19s in your voice: detected low voice energy...",
   "cues": ["Low energy", "Long pauses"]
}
```

Rule-based cue mapping (design specification):

Condition	Cue Tag
pitch_std_hz < 30	Monotone pitch
pitch_mean_hz < 150	Low vocal pitch
energy_mean < 0.015	Low energy
pause_ratio > 0.4	Long pauses
speech_rate < 0.5	Slow speech
pitch_std_hz > 80	Varied pitch
energy_mean > 0.04	Strong energy
speech_rate > 0.75	Fluent speech

Table 5.7

5.3.3.2 Subtype Profiling Subsystem (subtype.py)

Design type: Heuristic profile matcher (not supervised classifier).
Seven profiles: MDD, Dysthymia, Bipolar, SAD, Postpartum, Psychotic, Situational.
Mechanism:

1. Aggregate acoustic features across all segments (weighted mean).
2. Score each profile via Gaussian distance from idealized symptom vectors.
3. Apply optional context boosts from user checkboxes.
4. Softmax-normalize scores → probability distribution.
5. Return primary type, rankings, matched symptoms, disclaimer.

Design constraint: Subtype output is only generated when prediction == "Depressed" and is always accompanied by a non-diagnostic disclaimer.

5.3.3.3 API Serialization Subsystem (api_utils.py)

Transforms the internal predict() dictionary into a frontend-ready JSON payload:

- Renders 4–5 matplotlib charts → PNG → Base64 strings
- Rounds numeric values for display
- Structures timeline_explanations, feature_importance, subtype, charts

Response schema (top-level keys):

```
prediction, confidence, probability_depressed, audio_duration_sec,
n_segments, prediction_reason, key_segment, timeline_explanations,
segment_probabilities, segment_explanations, summary, acoustic_features,
feature_importance[], subtype{}, charts{spectrogram, grad_cam, timeline,
features, subtype}
```

5.3.3.4 Extended Module Architecture

```
segment_details + prediction
       │
       ├─► build_timeline_explanations() ──► timeline cards
       ├─► build_prediction_reason() ──────► summary text
       ├─► compute_grad_cam() ─────────────► heatmap array
       ├─► compute_feature_importance() ───► ranked features
       ├─► classify_subtype() ─────────────► subtype rankings
       │
       ▼
result_to_json() ──► Base64 charts + structured JSON ──► Web UI
```

5.4 Data Flow and Interaction Design

5.4.1 End-to-End Data Flow

```
┌──────────┐     upload      ┌──────────┐    temp file    ┌──────────────────┐
│  User    │────────────────►│ Frontend │────────────────►│ FastAPI server   │
│ (Browser)│                 │ app.js   │  POST /predict  │ server.py         │
└──────────┘                 └──────────┘                 └────────┬─────────┘
    ▲                                                              │
    │                                                              ▼
    │                                                   ┌──────────────────┐
    │                                                   │ DepressionPredictor│
    │                                                   │ predict.py        │
    │                                                   └──────┬───────────┘
    │         JSON + Base64 charts                            │
    │◄─────────────────────────────────────────────────────────┘
    │                                                              │
    │                                 ┌────────────────────────────┘
    │                                 │
```

```
   │   ┌─────────────────────────┼─────────────────────────┐
   │   ▼                         ▼                         ▼
   │  Preprocessing            ML Engine             Extended Module
   │  (load, segment,           (CNN, MLP,             (Grad-CAM, timeline,
   │   mel, features)            aggregation)           subtype, charts)
   │                              │                         │
   │                              └────────────┬────────────┘
   │                                           ▼
   │                                   result_to_json()
   │                                           │
   └───────────────────────────────────────────┘
```

5.4.2 Training Data Flow

```
depressed/ + non depressed/ folders
       │
       ▼
discover_audio_files() → (path, label, participant_id)
       │
       ▼
build_segment_dataset() → spectrograms[], labels[], metadata[]
       │
       ▼
participant_level_split() → train_idx[], test_idx[]
       │
       ├─► MelSpectrogramDataset + DataLoader → DepressionCNN training
       │
            └─► features_to_vector() + StandardScaler → FeatureClassifier
training
               │
               ▼
      models/depression_cnn.pt, feature_model.pt, feature_scaler.pkl
```

5.4.3 Interaction Design (Web UI)

The frontend (frontend/index.html, static/js/app.js, static/css/style.css) implements a task-oriented, tab-based interaction model.

5.4.3.1 User Journey

Step	User Action	System Response
1	Open http://localhost:8765	Health check (GET /api/health); status badge shows model readiness
2	Drag/drop or browse audio file	File preview displayed; Analyze button enabled

3	(Optional) Check context boxes	Flags stored for subtype boosting
4	Click "Analyze Recording"	Loading spinner; POST /api/predict with FormData
5	View results	Prediction badge (Depressed / Non-Depressed), confidence, duration
6	Navigate tabs	Six explanation views (see below)

Table 5.8

5.4.3.2 Results Tab Structure

Tab	Content	Design Intent
Why This Result?	prediction_reason + timeline cards with exact seconds	Primary explanation; answers "why"
Depression Type	Subtype rankings + disclaimer	Optional research exploration
Spectrogram	Full recording mel plot with colored regions	Spatial overview of where model looked
Voice Timeline	Bar chart of probability vs. time	Temporal pattern inspection
Grad-CAM	Heatmap on key segment with peak annotation	Model-attention evidence
Features	Horizontal bar chart of top 10 acoustic features	Semantic feature evidence

Table 5.9

5.4.3.3 Interaction Design Principles

Principle	Implementation
Progressive disclosure	Prediction shown first; details in tabs
Dual coding	Text explanations + visual charts for same evidence
Color semantics	Red (#e85d6f) = depression signal; Green (#3ecf8e) = non-depression
Explicit disclaimer	Visible on upload screen and in summary
Error feedback	HTTP 400/500/503 with user-readable messages
Accessibility of time	All times in seconds (e.g., "16s – 19s"), not frame indices

Table 5.10

5.4.3.4 API Interaction Contract

Request:

```
POST /api/predict
Content-Type: multipart/form-data
file: <audio binary>
chronic: bool (default false)
recent_stress: bool
postpartum: bool
seasonal: bool
mood_swings: bool
```

Response: 200 OK with JSON body (see Section 5.3.3.3).
Error cases:

Code	Condition
400	Unsupported format, empty file
503	Model not trained (depression_cnn.pt missing)
500	Inference failure

Table 5.11

5.5 Extension Integration into AI Framework

The extension integrates into the AI framework at three levels: model, pipeline, and application.

5.5.1 Model-Level Integration

The CNN backbone remains a standard PyTorch nn.Module. The extension attaches via:

1. Forward-hook on conv3 — captures activations and gradients for Grad-CAM without architectural change.
2. Parallel FeatureClassifier — independent model sharing labels but not weights with the CNN.
3. Post-inference aggregation — DepressionPredictor merges CNN probabilities, feature vectors, and explanation artifacts into a single result dictionary.

```
Standard CNN classifier
       │
       ├── [unchanged weights] ──▶ P(depressed)
       │
       └── [Grad-CAM hook on conv3] ──▶ spatial heatmap
```

```
                  +
      FeatureClassifier (parallel) ──▶ feature importance
                  │
                  ▼
      Explanation fusion layer (explain.py + predict.py)
```

5.5.2 Pipeline-Level Integration

The extension is inserted as a post-classification stage in the inference pipeline. This follows the post-hoc XAI pattern used in Captum, LIME, and SHAP toolkits, but domain-specialized for speech depression detection:

Pipeline Stage	Baseline System	Extended System
Preprocess	✓	✓ (identical)
Classify	✓ CNN only	✓ CNN (identical backbone)
Explain	✗	✓ Grad-CAM + features + timeline
Subtype	✗	✓ Profile matcher (optional)
Visualize	✗	✓ 5 chart types
Deploy	✗ CLI only	✓ FastAPI + Web UI

Table 5.12

5.5.3 Application-Level Integration

At the application layer, the extension conforms to Clinical Decision Support System (CDSS) architecture:

```
┌─────────────────────────────────────────────┐
│            CLINICIAN / RESEARCHER            │
│       (human decision-maker — external)      │
└─────────────────────┬───────────────────────┘
                     │ interprets
┌────────────────────▼────────────────────────┐
│          PRESENTATION (Web UI)                 │
│  prediction + explanations + disclaimers       │
└─────────────────────┬───────────────────────┘
                     │
┌────────────────────▼────────────────────────┐
│          REASONING (Extended Module)         │
│  Grad-CAM · timeline · feature importance    │
└─────────────────────┬───────────────────────┘
                     │
┌────────────────────▼────────────────────────┐
│          INFERENCE (ML Engine)               │
│  DepressionCNN + FeatureClassifier           │
```

```
└──────────────────────────────┬──────────────────────────────┘
                               │
┌──────────────────────────────▼──────────────────────────────┐
│            DATA (Preprocessing Module)                      │
│   normalized, segmented speech features                     │
└─────────────────────────────────────────────────────────────┘
```

5.5.4 Integration with External AI Ecosystem

Ecosystem Component	Integration Point
PyTorch	Model definition, training, Grad-CAM hooks, torch.save/load
scikit-learn	StandardScaler, accuracy_score, f1_score
Librosa	Industry-standard speech feature extraction
SHAP	Optional KernelExplainer (disabled in fast mode)
FastAPI / Uvicorn	Production-style REST serving
Matplotlib	Server-side chart rendering → Base64 embedding
DAIC-WOZ corpus	Training data format and label convention

Table 5.13

5.5.5 Extensibility Hooks

The design includes explicit extension points for future AI framework growth:

Hook	Location	Future Use
augment=True	build_segment_dataset()	SpecAugment, pitch shifting
fast=False	compute_feature_importance()	Full SHAP values
context dict	predict(), classify_subtype()	Additional clinical metadata
model_dir param	DepressionPredictor.__init__()	Multi-model serving
Alternative UI	app.py (Streamlit)	Rapid prototyping
CUDA device	torch.device("cuda")	GPU acceleration

Table 5.14

5.6 Summary

This chapter presented the analysis and design of the explainable speech-based depression detection extension. The design rationale—modularity, dual-path processing, temporal granularity, non-invasive explainability, and deployable decision support—directly addresses the research gap identified in Chapters 2 and 3.

The top-level architecture decomposes into three modules:

1. Preprocessing Module (features.py, data.py) — audio loading, 16 kHz normalization, 3-second segmentation, mel-spectrogram and 23-feature extraction, participant-level splitting.
2. ML Engine (model.py, train.py) — DepressionCNN for spectrogram classification, FeatureClassifier for acoustic attribution, training/evaluation pipeline producing persisted model artifacts.
3. Extended Module (explain.py, subtype.py, api_utils.py) — Grad-CAM, timeline explanations, subtype profiling, chart rendering, and JSON API serialization.

Data flow was specified for both training and inference paths, and the web UI interaction design was documented as a six-tab, progressive-disclosure interface with explicit ethical disclaimers.

The extension integrates into the AI framework at model level (Grad-CAM hooks on conv3), pipeline level (post-classification explanation stage), and application level (CDSS-style layered architecture). Extensibility hooks support future scaling to larger corpora, SHAP attribution, multimodal fusion, and GPU deployment.

Chapter 6 will present implementation details and evaluation results—translating this design into measured performance metrics, explanation outputs, and critical analysis of system behavior on the 9-participant DAIC-WOZ–style dataset.

CHAPTER 6
IMPLEMENTATION

6.1 Introduction

Chapter 5 defined the analysis and design of the explainable speech-based depression detection extension—module decomposition, data flows, API contracts, and integration points. This chapter describes how that design was realized in code: the development environment, hardware and software platforms, module-wise implementation, algorithms, workflow diagrams, component integration, and testing performed during development.

The implementation produces a fully functional pipeline spanning:

- Offline model training (train.py)
- Online inference with explainability (src/predict.py)
- REST API serving (server.py)
- Interactive web interface (frontend/)
- Alternative Streamlit prototype (app.py)

All source code is organized under /Users/ayeshaasarak/Desktop/Data/ with a modular src/ package, persisted model artifacts in models/, and DAIC-WOZ–style interview recordings in depressed/ and non depressed/ folders.

6.2 Overall Development Environment

6.2.1 Project Structure

The project follows a flat-root, modular-package layout:

```
Data/
├── depressed/                # 5 depressed participants (*_AUDIO.wav)
├── non depressed/            # 4 non-depressed participants
├── models/                   # Trained artifacts
│   ├── depression_cnn.pt
│   ├── feature_model.pt
│   ├── feature_scaler.pkl
│   └── training_metadata.json
├── src/
│   ├── config.py             # Constants and paths
│   ├── data.py               # Dataset loading and splitting
│   ├── features.py           # Audio preprocessing and features
│   ├── model.py              # CNN and MLP architectures
│   ├── explain.py            # Grad-CAM and visualizations
│   ├── subtype.py            # Subtype profile matcher
```

```
│   ├── predict.py          # Inference orchestrator
│   └── api_utils.py        # JSON/chart serialization
├── frontend/
│   ├── index.html
│   └── static/css/, js/
├── train.py                # Training entry point
├── server.py               # FastAPI server
├── app.py                  # Streamlit alternative UI
├── requirements.txt
└── start_server.sh
```

6.2.2 Development Methodology

Implementation followed an iterative, bottom-up approach:

Phase	Activities
Phase 1	Audio loading, segmentation, mel-spectrogram extraction (features.py)
Phase 2	Dataset discovery, participant-level split, PyTorch Dataset (data.py)
Phase 3	CNN and MLP model definitions (model.py)
Phase 4	Training loop, evaluation metrics, model persistence (train.py)
Phase 5	Grad-CAM, timeline explanations, chart plotting (explain.py)
Phase 6	End-to-end inference orchestration (predict.py)
Phase 7	FastAPI server, JSON serialization, web frontend (server.py, api_utils.py, frontend/)
Phase 8	Subtype profile matcher (subtype.py)
Phase 9	Integration testing on sample audio files

Table 6.1

6.2.3 Environment Configuration

A required environment variable prevents Numba caching errors with Librosa:

```
export NUMBA_CACHE_DIR="$(pwd)/.numba_cache"
```

Training:

```
cd /Users/ayeshaasarak/Desktop/Data
pip3 install -r requirements.txt
python3 train.py
```

Serving:

```
python3 -m uvicorn server:app --host 127.0.0.1 --port 8765 --reload
```

6.3 Hardware and Software Platforms

6.3.1 Hardware Platform

Component	Specification	Usage
Processor	CPU (Apple Silicon / x86_64)	Training and inference
GPU	Optional CUDA device	Auto-detected; not required
RAM	≥ 8 GB recommended	Model loading, spectrogram buffers
Storage	~50 MB for models + dataset	models/ (~1.5 MB), audio files
Microphone	Optional (user uploads)	Web UI accepts pre-recorded files

Table 6.2

The implementation auto-selects compute device:

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

All development and testing were performed on CPU, demonstrating deployability without specialized GPU hardware.

6.3.2 Software Platform

Category	Technology	Version
OS	macOS (darwin 25.5.0)	—
Language	Python	3.9+
Deep Learning	PyTorch	≥ 2.0
Audio	Librosa, SoundFile	≥ 0.10, ≥ 0.12
ML Utilities	scikit-learn, NumPy, Pandas	≥ 1.3, ≥ 1.24, ≥ 2.0
Explainability	Custom Grad-CAM; SHAP (optional)	≥ 0.43
Visualization	Matplotlib, Pillow	≥ 3.7, ≥ 10.0
Web Backend	FastAPI, Uvicorn	≥ 0.104, ≥ 0.24
Web Frontend	HTML5, CSS3, JavaScript (vanilla)	—
Prototype UI	Streamlit	≥ 1.28
Upload Handling	python-multipart	≥ 0.0.6

Table 6.3

6.3.3 Dependency Management

All dependencies are declared in requirements.txt and installed via pip:

```
torch>=2.0.0

librosa>=0.10.0
```

```
scikit-learn>=1.3.0
fastapi>=0.104.0
uvicorn>=0.24.0
matplotlib>=3.7.0
shap>=0.43.0
streamlit>=1.28.0
...
```

6.4 Moodule wise Implementation

6.4.1 Configuration Module (src/config.py)

Centralizes all hyperparameters and file paths.
Key constants implemented:

Constant	Value	Purpose
SAMPLE_RATE	16000	Audio resampling rate
N_MELS	128	Mel filterbank bands
N_MFCC	13	MFCC coefficients
HOP_LENGTH	512	STFT hop (32 ms at 16 kHz)
N_FFT	2048	FFT window size
SEGMENT_DURATION	3.0 s	Segmentation window
SEGMENT_OVERLAP	0.5	50% overlap between windows
FEATURE_NAMES	23 entries	Ordered acoustic feature list
MODEL_PATH	models/depression_cnn.pt	CNN checkpoint path

Table 6.4

6.4.2 Features Module (src/features.py)

Implements the Preprocessing Module from Chapter 5.

```
load_audio(path, max_duration)
```

- Loads via librosa.load(sr=16000, mono=True)
- Trims silence: librosa.effects.trim(top_db=25)
- Peak-normalizes to unit amplitude
- Returns np.float32 waveform

```
segment_audio_with_times(y, duration, overlap)
```

- Computes seg_len = duration × SAMPLE_RATE
- Hop size: seg_len × (1 - overlap) → 1.5 s hop for 3 s windows

- Returns list of {"audio", "start_sec", "end_sec"} dicts
- Handles short audio via zero-padding

`compute_mel_spectrogram(y)`

- librosa.feature.melspectrogram(n_fft=2048, hop_length=512, n_mels=128)
- Log-power conversion: librosa.power_to_db()
- Per-segment z-score normalization
- Output shape: (1, 128, T)

`extract_acoustic_features(y)`

- Pitch: librosa.piptrack() — per-frame max pitch, then mean/std
- Energy: librosa.feature.rms()
- Speech rate / pause ratio: frames above 20th-percentile RMS threshold
- Spectral: centroid, rolloff, bandwidth, zero-crossing rate
- MFCCs: 13 coefficients, mean per segment
- Returns dict of 23 named floats

augment_audio(y) (optional, disabled in default training)

- Random choice: time-stretch, pitch-shift, additive noise, or no-op

6.4.3 Data Module (src/data.py)

`discover_audio_files(data_dir)`

- Scans depressed/ and non depressed/ participant folders
- Locates *_AUDIO.wav per participant
- Returns (path, label, participant_id) tuples
- Labels: 1 = depressed, 0 = non-depressed

`build_segment_dataset(...)`

- Iterates all audio files
- Segments, computes mel specs and features
- Optional subsampling: `max_segments_per_file=12`, `max_duration=60s`
- Returns `(spectrograms, labels, metadata)` — 108 segments total

`MelSpectrogramDataset`

- PyTorch `Dataset` wrapping spectrogram arrays and float labels
- `__getitem__` returns `(torch.Tensor, torch.Tensor)`

`participant_level_split(metadata, test_ratio=0.22)`

- Groups segment indices by `participant_id`
- Randomly selects ~22% of participants for test (seed 42)
- Prevents speaker leakage between train and test
-

6.4.4 Model Module (src/model.py)

`DepressionCNN(nn.Module)`

Implemented layers:

Implemented layers:

```
self.conv1 = nn.Conv2d(1, 32, kernel_size=3, padding=1)

self.bn1   = nn.BatchNorm2d(32)
```

```
self.conv2 = nn.Conv2d(32, 64, kernel_size=3, padding=1)
self.bn2   = nn.BatchNorm2d(64)
self.conv3 = nn.Conv2d(64, 128, kernel_size=3, padding=1)  # Grad-CAM target
self.bn3   = nn.BatchNorm2d(128)
self.pool  = nn.MaxPool2d(2, 2)
self.global_pool = nn.AdaptiveAvgPool2d((4, 4))
self.fc1   = nn.Linear(2048, 128)
self.fc2   = nn.Linear(128, 1)
self.dropout = nn.Dropout(0.4)
```

Grad-CAM support:

- `_grad_cam_activations` stored after `conv3`
- `_save_gradients` backward hook on `conv3` output
- `predict_proba(x)` helper for single-sample inference

```
FeatureClassifier(nn.Module)
nn.Sequential(
   nn.Linear(n_features, 64), nn.ReLU(), nn.Dropout(0.3),
   nn.Linear(64, 32), nn.ReLU(),
   nn.Linear(32, 1)
)
```

6.4.5 Training Module (train.py)

Training loop implementation:

Step	Implementation Detail
Data load	build_segment_dataset(augment=False, max_duration=60, max_segments=12)
Split	participant_level_split(test_ratio=0.22)
CNN train	12 epochs, batch=8, Adam lr=1e-3, weight_decay=1e-4, BCEWithLogitsLoss
Evaluation	accuracy_score, precision_score, recall_score, f1_score
Deploy fine-tune	Copy best weights → 8 epochs on full 108 segments, lr=5e-4
Feature train	StandardScaler.fit_transform → 30 epochs, lr=1e-3
Save	torch.save, pickle.dump(scaler), save_training_metadata()

Table 6.5

Saved test metrics (models/training_metadata.json):

Metric	Value

Participants	9
Segments	108
Accuracy	0.75
Precision	1.00
Recall	0.75
F1-Score	0.857

Table 6.6

6.4.6 Explainability Module (src/explain.py)

```
compute_grad_cam(model, spectrogram)
```

1. Forward pass with requires_grad=True
2. Backward pass on logit
3. Weight activations by mean pooled gradients
4. ReLU + bilinear upsample to input size
5. Min-max normalize to [0, 1]

```
build_timeline_explanations(segment_details, prediction, top_n=6)
```

- Ranks segments by probability (supporting vs. opposing)
- Calls explain_voice_at_time() for natural-language text
- Extracts symptom cue tags via _extract_cues()
- Sorts output chronologically by start_sec

```
explain_voice_at_time(start, end, prob, features, prediction)
```

- Rule-based mapping of acoustic thresholds to human-readable cues
- Returns sentence: "At 16s – 19s in your voice: detected low voice energy, long pauses..."

Visualization functions:

- plot_full_spectrogram() — full recording with colored region overlays
- plot_grad_cam() — side-by-side spectrogram + CAM with peak-second annotation
- plot_timeline_chart() — bar chart of probability vs. time

```
compute_feature_importance(feature_model, vectors, fast=True)
```

- Fast mode: gradient × input, mean absolute value per feature
- Slow mode: SHAP KernelExplainer (optional, disabled by default)

6.4.7 Subtype Module (src/subtype.py)

```
classify_subtype(segment_details, prediction, depression_prob, context)
```

Implementation steps:

1. Return "Not Applicable" if prediction is Non-Depressed
2. Aggregate features across depressed segments (mean)

3. Score each of 7 subtypes via Gaussian profile matching:

```
score = exp(-0.5 * (|actual - ideal| / tolerance)²) × weight
```

4. Apply context boosts (chronic → Dysthymia +0.25, etc.)
5. Scale by depression probability: score × (0.6 + 0.4 × prob)
6. Softmax normalization across subtypes
7. Return rankings, matched symptoms, acoustic summary, disclaimer

6.4.8 Inference Module (src/predict.py)

DepressionPredictor class — central orchestrator.

```
__init__(model_dir)
```

- Loads `depression_cnn.pt` → `DepressionCNN`
- Loads `feature_model.pt` → `FeatureClassifier`
- Loads `feature_scaler.pkl` → `StandardScaler`
- Sets `eval()` mode on both models

predict(audio_path, context) — 10-step pipeline:

1. Load audio (max 180 s)
2. Segment with timestamps
3. Compute full mel spectrogram
4. Per segment: mel spec → CNN probability; extract 23 features
5. Aggregate: mean probability → prediction + confidence
6. Build timeline explanations and prediction reason
7. Select key segment for Grad-CAM (highest-impact supporting segment)
8. Compute Grad-CAM heatmap
9. Compute feature importance across all segments
10. Classify subtype (if depressed); return full result dict

6.4.9 API Module (server.py + src/api_utils.py)

server.py — FastAPI endpoints:

Endpoint	Method	Implementation
/	GET	Serves frontend/index.html
/api/health	GET	Checks MODEL_PATH.exists()
/api/predict	POST	Accepts audio file + 5 context Form booleans
/static/*	GET	StaticFiles mount for CSS/JS

Table 6.7

Upload handling:

- Validates extension: .wav, .mp3, .flac, .ogg, .m4a, .webm
- Writes to tempfile.NamedTemporaryFile
- Calls DepressionPredictor.predict()
- Deletes temp file in finally block

api_utils.result_to_json(result)

- Renders 4–5 matplotlib figures
- Converts to Base64 PNG strings via fig_to_base64()
- Returns structured JSON for frontend consumption

6.4.10 Frontend Module (frontend/)

index.html — single-page application with:

- Drag-and-drop upload zone
- Optional context checkboxes
- Results section with 6 tab panels
- Status badge (model ready / not trained)
- Ethical disclaimer banner

static/js/app.js — client-side logic:

- checkHealth() on page load → GET /api/health
- File selection via drag-drop or browse
- analyzeBtn → POST /api/predict with FormData
- Tab switching, chart rendering from Base64, timeline card population
- Toast notifications for errors

static/css/style.css — dark-theme UI with:

- Depression red (#e85d6f), non-depression green (#3ecf8e)
- Card-based layout, responsive grid

6.4.11 Alternative UI (app.py)

Streamlit prototype providing a simpler upload-and-predict interface for rapid demonstration without the full web frontend. Shares the same DepressionPredictor backend.

6.5 Algorithms and Pseudocode

6.5.1 Algorithm 1: Audio Preprocessing and Segmentation

```
ALGORITHM: PreprocessAudio(audio_path, max_duration)
INPUT:  Raw audio file path, optional duration cap
OUTPUT: List of segment dicts {audio, start_sec, end_sec}
1.  y ← LOAD(audio_path, sr=16000, mono=True, duration=max_duration)
2.  y ← TRIM_SILENCE(y, top_db=25)
3.  IF max(|y|) > 0 THEN y ← y / max(|y|)
4.  seg_len ← 3.0 × 16000
5.  hop ← seg_len × 0.5
6.  segments ← []
7.  FOR start ← 0 TO len(y) - seg_len STEP hop DO
8.      start_sec ← start / 16000
9.      end_sec   ← (start + seg_len) / 16000
```

```
10.        APPEND {audio: y[start:start+seg_len], start_sec, end_sec} TO segments
11. RETURN segments
```

6.5.2 Algorithm 2: Mel-Spectrogram Feature Extraction

```
ALGORITHM: ExtractMelSpectrogram(segment_audio)
INPUT:  3-second waveform array
OUTPUT: Normalized log-mel tensor (1, 128, T)
1.  mel ← MELSPECTROGRAM(y, sr=16000, n_fft=2048, hop=512, n_mels=128)
2.  log_mel ← POWER_TO_DB(mel)
3.  log_mel ← (log_mel - MEAN(log_mel)) / (STD(log_mel) + ε)
4.  RETURN log_mel reshaped to (1, 128, T)
```

6.5.3 Algorithm 3: CNN Training

```
ALGORITHM: TrainDepressionCNN(train_loader, test_loader, epochs=12)
INPUT:  PyTorch DataLoaders, hyperparameters
OUTPUT: Trained DepressionCNN weights, evaluation metrics
1.  model ← DepressionCNN()
2.  optimizer ← Adam(lr=1e-3, weight_decay=1e-4)
3.  criterion ← BCEWithLogitsLoss()
4.  FOR epoch ← 1 TO epochs DO
5.      FOR each batch (x, y) IN train_loader DO
6.          logits ← model(x)
7.          loss ← criterion(logits, y)
8.          BACKPROPAGATE and UPDATE weights
9.      IF epoch MOD 4 = 0 THEN
10.         metrics ← EVALUATE(model, test_loader)
11. metrics ← EVALUATE(model, test_loader)
12. RETURN model, metrics
```

6.5.4 Algorithm 4: Inference and Aggregation

```
ALGORITHM: PredictDepression(audio_path)
INPUT:  Audio file path, optional context flags
OUTPUT: Prediction dict with explanations
1.  segments ← PreprocessAudio(audio_path)
2.  segment_details ← []
3.  FOR each seg IN segments DO
4.      spec ← ExtractMelSpectrogram(seg.audio)
5.      prob ← SIGMOID(CNN(spec))
6.      feats ← ExtractAcousticFeatures(seg.audio)
7.      APPEND {start_sec, end_sec, prob, features: feats}
8.  avg_prob ← MEAN(all probs)
9.  prediction ← "Depressed" IF avg_prob ≥ 0.5 ELSE "Non-Depressed"
```

```
10. confidence ← avg_prob IF depressed ELSE (1 - avg_prob)
11. timeline ← BuildTimelineExplanations(segment_details, prediction)
12. key_seg ← SELECT highest-impact supporting segment
13. cam ← ComputeGradCAM(CNN, key_seg.spectrogram)
14. importance ← ComputeFeatureImportance(FeatureMLP, all feature vectors)
15. subtype ← ClassifySubtype(segment_details, prediction, context)
16. RETURN {prediction, confidence, timeline, cam, importance, subtype, ...}
```

6.5.5 Algorithm 5: Grad-CAM

```
ALGORITHM: ComputeGradCAM(model, spectrogram)
INPUT:  Trained CNN, single mel-spectrogram
OUTPUT: 2D importance heatmap (128 × T)
1.  x ← spectrogram with requires_grad = True
2.  logit ← model.forward(x)
3.  model.zero_grad()
4.  logit.backward()
5.  A ← model._grad_cam_activations        // (1, 128, h, w)
6.  G ← model._grad_cam_gradients          // (1, 128, h, w)
7.  weights ← MEAN(G, dims=(2,3), keepdim=True)
8.  cam ← RELU(SUM(weights × A, dim=1))
9.  cam ← BILINEAR_UPSAMPLE(cam, size=input_shape)
10. cam ← NORMALIZE(cam, min=0, max=1)
11. RETURN cam
```

6.5.6 Algorithm 6: Participant-Level Split

```
ALGORITHM: ParticipantLevelSplit(metadata, test_ratio=0.22)
INPUT:  Segment metadata with participant_id, split ratio
OUTPUT: train_indices[], test_indices[]
1.  groups ← GROUP metadata indices BY participant_id
2.  pids ← SORTED(unique participant_ids)
3.  n_test ← MAX(1, FLOOR(len(pids) × test_ratio))
4.  test_pids ← RANDOM_CHOICE(pids, n_test, seed=42)
5.  FOR each pid, indices IN groups DO
6.      IF pid IN test_pids THEN test_idx.extend(indices)
7.      ELSE train_idx.extend(indices)
8.  RETURN train_idx, test_idx
```

6.5.7 Algorithm 7: Subtype Profile Matching

```
ALGORITHM: ClassifySubtype(segment_details, prediction, context)
INPUT:  Segment details, prediction label, optional context
OUTPUT: Subtype rankings with probabilities
1.  IF prediction ≠ "Depressed" THEN RETURN not_applicable
2.  agg ← MEAN(features) across segments where prob ≥ 0.5
```

```
3.   FOR each subtype IN {MDD, Dysthymia, Bipolar, SAD, Postpartum, Psychotic, Situational} DO
4.     score ← 0
5.         FOR each (feature, ideal, tolerance, weight) IN ACOUSTIC_PROFILES[subtype] DO
6.         score += GAUSSIAN_SCORE(agg[feature], ideal, tolerance) × weight
7.     raw_scores[subtype] ← score / weight_sum
8.  APPLY context boosts from user checkboxes
9.  SCALE scores by depression probability
10. probs ← SOFTMAX(raw_scores)
11. RETURN sorted rankings, primary type, matched symptoms, disclaimer
```

6.6 Workflow Diagrams / Flowcharts

6.6.1 System Implementation Flowchart

```
                ┌──────────────────┐
                │   START SYSTEM   │
                └─────────┬────────┘
                          │
          ┌───────────────┴───────────────┐
          │                               │
          ▼                               ▼
┌──────────────────┐            ┌──────────────────┐
│  OFFLINE TRAIN   │            │  ONLINE SERVE    │
│  python train.py │            │  uvicorn server  │
└─────────┬────────┘            └─────────┬────────┘
          │                               │
          ▼                               ▼
┌──────────────────┐            ┌──────────────────┐
│ Discover audio   │            │ User opens UI    │
│ 9 participants   │            │ GET /api/health  │
└─────────┬────────┘            └─────────┬────────┘
          │                               │
          ▼                               ▼
┌──────────────────┐            ┌──────────────────┐
│ Preprocess +     │            │ Upload audio     │
│ segment (108)    │            │ POST /api/predict│
└─────────┬────────┘            └─────────┬────────┘
          │                               │
          ▼                               ▼
┌──────────────────┐            ┌──────────────────┐
│ Participant      │            │ DepressionPredictor│
```

```
│ level split        │          │ .predict()         │
└─────────┬──────────┘          └─────────┬──────────┘
          │                               │
          ▼                               ▼
┌────────────────────┐          ┌────────────────────┐
│ Train CNN 12 ep    │          │ JSON + charts      │
│ Eval on test       │          │ Render in UI       │
└─────────┬──────────┘          └─────────┬──────────┘
          │                               │
          ▼                               ▼
┌────────────────────┐          ┌────────────────────┐
│ Fine-tune 8 ep     │          │   DISPLAY RESULTS  │
│ Train feature MLP  │          │   (6 tabs)         │
└─────────┬──────────┘          └────────────────────┘
          │
          ▼
┌────────────────────┐
│ Save models/       │
│ metadata.json      │
└────────────────────┘
```

6.6.2 Inference Pipeline Flowchart

```
Audio Upload
    │
    ▼
┌─────────────┐     No       ┌──────────────┐
│ Valid       │─────────────►│ Return 400   │
│ format?     │              │ Error        │
└──────┬──────┘              └──────────────┘
       │ Yes
       ▼
┌─────────────┐     No       ┌──────────────┐
│ Model       │─────────────►│ Return 503   │
│ exists?     │              │ Not trained  │
└──────┬──────┘              └──────────────┘
       │ Yes
       ▼
┌─────────────────────────────────────────┐
│ load_audio → segment (3s, 50% overlap) │
└────────────────┬────────────────────────┘
                 │
                 ▼
```

```
┌────────────────────────────────────────┐
│ FOR EACH SEGMENT:                       │
│   mel spec → CNN → prob                │
│   waveform → 23 acoustic features     │
└────────────────┬───────────────────────┘
                 │
                 ▼
┌────────────────────────────────────────┐
│ avg_prob = mean(probs)                  │
│ prediction = Depressed if avg ≥ 0.5    │
└────────────────┬───────────────────────┘
                 │
       ┌─────────┴─────────┐
       ▼                   ▼
┌─────────────┐    ┌──────────────────┐
│ Timeline     │    │ Grad-CAM on key  │
│ explanations │    │ segment          │
└──────┬──────┘    └────────┬─────────┘
       │                    │
       └────────┬───────────┘
                ▼
┌────────────────────────────────────────┐
│ Feature importance + subtype matching   │
└────────────────┬───────────────────────┘
                 │
                 ▼
┌────────────────────────────────────────┐
│ result_to_json() → charts as Base64    │
│ Return 200 OK to frontend               │
└────────────────────────────────────────┘
```

6.6.3 Grad-CAM Computation Flowchart

```
Input: mel spectrogram tensor
        │
        ▼
  Forward pass (requires_grad=True)
        │
        ▼
  Store conv3 activations (A)
        │
        ▼
  Compute logit → scalar output
```

```
       |
       ▼
  Backward pass → gradients (G)
       |
       ▼
  weights = mean(G, spatial dims)
       |
       ▼
  cam = ReLU(Σ weights_k × A_k)
       |
       ▼
  Upsample to input dimensions
       |
       ▼
  Normalize [0, 1] → heatmap
       |
       ▼
  Overlay on spectrogram + annotate peak second
```

6.6.4 Web UI Interaction Flowchart

```
Page Load
   |
   ▼
checkHealth() ──► model_ready? ──No──► Show warning toast
   | Yes
   ▼
Display "Model ready" badge
   |
   ▼
User selects audio file (drag or browse)
   |
   ▼
Enable "Analyze Recording" button
   |
   ▼
User clicks Analyze
   |
   ▼
Show loading spinner
   |
   ▼
POST /api/predict (file + context flags)
```

```
  │
  ├── Error ──▶ Show toast, hide spinner
  │
  └── Success
        │
        ▼
  Display prediction badge + confidence
        │
        ▼
  Populate 6 tabs:
  Why | Subtype | Spectrogram | Timeline | Grad-CAM | Features
        │
        ▼
  User switches tabs to inspect explanations
```

6.7 Integration of Components

6.7.1 Import Dependency Graph

```
train.py
 ├── src.data      (build_segment_dataset, participant_level_split)
 ├── src.features  (features_to_vector, segment_audio, ...)
 └── src.model     (DepressionCNN, FeatureClassifier)
server.py
 ├── src.predict   (DepressionPredictor)
 └── src.api_utils (result_to_json)
src/predict.py
 ├── src.config
 ├── src.features
 ├── src.model
 ├── src.explain
 └── src.subtype
src/explain.py
 ├── src.config
 └── src.model
src/api_utils.py
 └── src.explain   (plot functions)
frontend/app.js
 └── HTTP → server.py (no direct Python imports)
```

6.7.2 Runtime Integration Sequence

Step	Component A	Component B	Interface

1	`app.js`	`server.py`	`POST /api/predict` (multipart)
2	`server.py`	`DepressionPredictor`	`predict(tmp_path, context)`
3	`predict.py`	`features.py`	`load_audio`, `segment_audio_with_times`, `compute_mel_spectrogram`, `extract_acoustic_features`
4	`predict.py`	`model.py`	`DepressionCNN.forward()`, `FeatureClassifier.forward()`
5	`predict.py`	`explain.py`	`compute_grad_cam`, `build_timeline_explanations`, `build_prediction_reason`
6	`predict.py`	`subtype.py`	`classify_subtype(segment_details, ...)`
7	`server.py`	`api_utils.py`	`result_to_json(result)`
8	`api_utils.py`	`frontend`	JSON with Base64 chart images
9	`app.js`	DOM	Render tabs, charts, timeline cards

Table 6.8

6.7.3 Model Artifact Integration

```
train.py
   |
   ├── saves → models/depression_cnn.pt
   ├── saves → models/feature_model.pt
   ├── saves → models/feature_scaler.pkl
   └── saves → models/training_metadata.json
                 |
                 ▼
DepressionPredictor.__init__()
   |
   ├── loads depression_cnn.pt  → self.cnn
   ├── loads feature_model.pt → self.feat_model
   └── loads feature_scaler.pkl → self.scaler
```

6.7.4 Shared Data Contract

The segment_details list is the central data structure connecting ML Engine and Extended Module:

```
segment_details = [
```

```
  {
      "start_sec": 16.0,
      "end_sec": 19.0,
      "prob": 0.71,
      "features": {
          "pitch_mean_hz": 142.3,
          "pitch_std_hz": 22.1,
          "energy_mean": 0.009,
          "pause_ratio": 0.45,
          ...
      }
  },
  ...
]
```

This structure is consumed by:

- build_timeline_explanations() — temporal NL explanations
- build_prediction_reason() — summary text
- compute_grad_cam() — via key segment selection
- classify_subtype() — feature aggregation
- plot_timeline_chart() — probability bar chart
- plot_full_spectrogram() — region highlighting

6.7.5 Error Handling Integration

Layer	Error	Handling
Frontend	No file selected	Analyze button disabled
Frontend	Server offline	Status badge + toast
API	Unsupported format	HTTP 400
API	Empty upload	HTTP 400
API	Model not found	HTTP 503 with instruction
API	Inference exception	HTTP 500 with message
Inference	Short audio (< 0.5 s)	Zero-filled features returned
Grad-CAM	Missing activations	Zero heatmap fallback

Table 6.9

6.8 Testing During Development

6.8.1 Unit-Level Testing

Informal unit testing was performed during development on individual functions:

Module	Test	Expected Behavior	Result

load_audio()	Load 319_AUDIO.wav	16 kHz mono, normalized, trimmed	✓ Pass
segment_audio_with_times()	60 s audio	~39 overlapping 3 s segments	✓ Pass
compute_mel_spectrogram()	3 s segment	Shape (1, 128, T), finite values	✓ Pass
extract_acoustic_features()	Depressed segment	Low energy, high pause ratio	✓ Pass
participant_level_split()	9 participants	No participant in both splits	✓ Pass
DepressionCNN.forward()	Batch of 8	Output shape (8, 1) logits	✓ Pass
compute_grad_cam()	Single spectrogram	Heatmap shape matches input	✓ Pass
classify_subtype()	Non-depressed prediction	Returns applicable: false	✓ Pass

Table 6.9

6.8.2 Training Validation

Check	Method	Outcome
Dataset size	print(len(labels))	108 segments confirmed
Class balance	Count labels	60 depressed, 48 non-depressed
Loss convergence	Monitor epoch loss	Decreasing loss over 12 epochs
Test metrics	`evaluate()` on held-out participants	Accuracy=0.75, F1=0.857
Model persistence	Reload saved checkpoint	Weights load without error
Metadata export	Read `training_metadata.json`	Valid JSON with metrics

Table 6.10

6.8.3 Inference Testing

Sample predictions on full audio files (deployed model):

File	True Label	Predicted	Confidence

depressed/319_AUDIO.wav	Depressed	Depressed	~54%
depressed/325_AUDIO.wav	Depressed	Depressed	~71%
non depressed/303_AUDIO.wav	Non-Depressed	Non-Depressed	~74%
non depressed/313_AUDIO.wav	Non-Depressed	Non-Depressed	~61%

Table 6.11

All four sample files classified correctly. Confidence values vary (54%–74%), indicating the model expresses uncertainty rather than uniform overconfidence.

6.8.4 API Testing

Test Case	Endpoint	Expected	Result
Health check (model trained)	GET /api/health	model_ready: true	✓ Pass
Health check (no model)	GET /api/health	model_ready: false	✓ Pass
Valid WAV upload	POST /api/predict	200 + JSON with charts	✓ Pass
Invalid .txt upload	POST /api/predict	400 error	✓ Pass
Empty file	POST /api/predict	400 error	✓ Pass
Context flags	POST /api/predict with checkboxes	Subtype boosts applied	✓ Pass
Temp file cleanup	After prediction	Temp file deleted	✓ Pass

Table 6.12

6.8.5 Frontend Testing

Test Case	Expected	Result
Page load health check	Status badge shows "Model ready"	✓ Pass
Drag-and-drop upload	File preview shown, button enabled	✓ Pass
Analyze without file	Button remains disabled	✓ Pass
Results display	Prediction badge, confidence, 6 tabs populated	✓ Pass
Tab switching	Correct panel shown per tab	✓ Pass
Chart rendering	Base64 images display in img elements	✓ Pass

Timeline cards	Time labels and cue tags visible	✓ Pass
Error toast	Shown on server/network failure	✓ Pass

Table 6.13

6.8.6 Explainability Validation (Qualitative)

Check	Observation
Grad-CAM peak time	Annotated in seconds on spectrogram plot
Timeline cues on depressed audio	Low energy, long pauses, monotone pitch detected
Timeline cues on non-depressed audio	Active rhythm, typical energy reported
Opposing segments	Shown alongside supporting evidence
Feature importance ranking	Energy and pause features rank highly on depressed samples
Disclaimer present	Visible in UI and summary output

Table 6.14

6.8.7 Known Limitations Identified During Testing

Issue	Impact	Mitigation
Small test set (2 participants)	Metrics may not generalize	Documented as preliminary
Segment correlation	Inflated effective sample size	Participant-level split applied
CPU-only training	Slower epochs	Acceptable for 108 segments
SHAP disabled in fast mode	Less rigorous feature attribution	Gradient importance used instead
Subtype module heuristic	Not clinically validated	Disclaimer enforced

Table 6.15

6.9 Summary

This chapter documented the complete implementation of the explainable speech-based depression detection extension.
The development environment was established on macOS with Python 3.9+, organized into a modular src/ package with clear separation between preprocessing (features.py, data.py), ML engine (model.py, train.py), explainability (explain.py, subtype.py), inference orchestration (predict.py), and deployment (server.py, api_utils.py, frontend/).
Seven core algorithms were specified with pseudocode: audio preprocessing, mel-spectrogram extraction, CNN training, inference aggregation, Grad-CAM, participant-level splitting, and subtype profile matching. Workflow diagrams illustrated the offline training path, online inference pipeline, Grad-CAM computation, and web UI interaction flow.
Component integration was achieved through a shared segment_details data contract, lazy model loading, REST API boundaries, and Base64 chart serializationTesting during development covered unit functions, training convergence, held-out evaluation (Accuracy=0.75, F1=0.857), sample-file inference, API error handling, frontend interaction, and qualitative explainability validation.
Chapter 7 will present results, evaluation, and discussion—analyzing system performance, explanation quality, limitations, and comparison with existing approaches.

CHAPTER 7
EVALUATION

7.1 Introduction

Chapters 5 and 6 described the design and implementation of the explainable speech-based depression detection extension. This chapter presents how the system was evaluated, the experimental conditions under which tests were conducted, and a critical analysis of the results obtained.

Evaluation addresses the three research questions established in Chapter 3:

RQ	Question
RQ1	Can a CNN trained on mel-spectrograms reliably classify depressive vs. non-depressive speech under participant-level splitting?
RQ2	Can XAI methods identify time regions and acoustic properties consistent with known depressive speech markers?
RQ3	Does integrating explainability degrade classification performance relative to the same CNN backbone?

Table 7.1

The evaluation combines quantitative classification metrics (Accuracy, Precision, Recall, F1-Score) on a held-out test participant with qualitative explainability analysis (Grad-CAM, timeline cues, feature importance). Full-recording inference on all nine participants is also reported to assess deployed-model behavior. Results are interpreted honestly, acknowledging the small dataset and the limitations of segment-level evaluation.

7.2 Evaluation Strategy

7.2.1 Evaluation Objectives

The evaluation strategy was designed to assess four dimensions:

1. Classification performance — Can the CNN distinguish depressed from non-depressed speech?
2. Generalization integrity — Are results free from speaker-level data leakage?
3. Explanation quality — Do XAI outputs align with clinically recognized acoustic markers?
4. Deployment viability — Does the full pipeline produce correct, interpretable results on complete interview recordings?

7.2.2 Evaluation Levels

Two evaluation granularities were applied:

Level	Unit of analysis	Purpose
Segment-level	Each 3-second overlapping window	Primary metric computation during training; matches literature practice
Recording-level (participant-level)	One prediction per full *_AUDIO.wav file	Stricter, clinically meaningful unit; mean probability across all segments

Table 7.2

7.2.3 Evaluation Protocol

```
┌──────────────────────────────────────────────────────────────┐
│  PHASE 1: Held-out participant evaluation (training)          │
│  • Participant-level split (22% holdout, seed=42)            │
│  • Train CNN for 12 epochs on 8 participants (96 segments)   │
│  • Evaluate on 1 held-out participant (12 segments)        │
│  • Metrics: Accuracy, Precision, Recall, F1                  │
└─────────────────────────────┬────────────────────────────────┘
                              ▼
┌──────────────────────────────────────────────────────────────┐
│  PHASE 2: Deployment model evaluation (full data)           │
│  • Fine-tune best CNN weights for 8 epochs on all 108 segs│
│  • Run full-file inference on all 9 participants             │
│  • Recording-level accuracy + confidence analysis            │
└─────────────────────────────┬────────────────────────────────┘
                              ▼
┌──────────────────────────────────────────────────────────────┐
│  PHASE 3: Explainability qualitative evaluation             │
│  • Grad-CAM peak-time annotation                             │
│  • Timeline cue inspection (depressed vs. non-depressed)   │
│  • Feature importance ranking consistency                    │
│  • Web UI functional testing                                 │
└──────────────────────────────────────────────────────────────┘
```

7.2.4 Evaluation Constraints

The following constraints were deliberately accepted and documented:

- No cross-dataset validation — DAIC-WOZ–style subset only
- No clinical expert rating of explanation quality (future work)
- No SHAP-based attribution in production — gradient importance used for speed

- Single held-out test participant — limits statistical power of test metrics

7.3 Experimental Setup

7.3.1 Hardware Configuration

Component	Specification
Operating System	macOS (darwin 25.5.0)
Processor	CPU (CUDA not used)
Device selection	torch.device("cpu")
RAM	Standard workstation (≥ 8 GB)

Table 7.3

All experiments were conducted on CPU to demonstrate that the system does not require GPU infrastructure for thesis-scale workloads.

7.3.2 Software Configuration

Component	Version / Setting
Python	3.9+
PyTorch	≥ 2.0
Librosa	≥ 0.10
scikit-learn	≥ 1.3
Random seed	42 (participant split)
Numba cache	NUMBA_CACHE_DIR=.numba_cache

Table 7.4

7.3.3 Training Hyperparameters

Parameter	Value
Sample rate	16,000 Hz
Segment duration	3.0 seconds
Segment overlap	50% (1.5 s hop)
Max training duration per file	60 seconds
Max segments per file (training)	12
Mel bands	128
FFT size	2048
Hop length	512

CNN epochs	12
Deploy fine-tune epochs	8 (full dataset)
Feature MLP epochs	30
Batch size	8
Optimizer	Adam
Learning rate (CNN)	1×10^{-3}
Learning rate (deploy fine-tune)	5×10^{-4}
Weight decay	1×10^{-4}
Loss function	BCEWithLogitsLoss
Dropout (CNN)	0.4
Decision threshold	0.5
Test split ratio	22% (participant-level)

Table 7.5

7.3.4 Inference Configuration

Parameter	Training	Inference (Web App)
Max audio duration	60 s	180 s
Max segments per file	12 (subsampled)	All overlapping windows
Aggregation	Mean segment probability	Mean segment probability
XAI mode	N/A	Grad-CAM + gradient feature importance

Table 7.6

7.4 Datasets and Test Cases

7.4.1 Dataset Source

The dataset follows the DAIC-WOZ (Distress Analysis Interview Corpus — Wizard-of-Oz) format:

- Clinical interviews between a participant and virtual agent "Ellie"
- Depression labels validated by PHQ-8 clinical assessments
- Binary classification: Depressed (PHQ-8 $\geq$ 10) vs. Non-Depressed (PHQ-8 $<$ 10)

7.4.2 Dataset Composition

Category	Participant IDs	Audio Files	Segments (train config)
Depressed	319, 320, 321, 325, 329	5 × *_AUDIO.wav	60 (12 each)
Non-Depressed	303, 304, 312, 313	4 × *_AUDIO.wav	48 (12 each)
Total	9	9 interviews	108 segments

Table 7.7

Each participant folder also contains auxiliary files (*_COVAREP.csv, *_FORMANT.csv, *_TRANSCRIPT.csv, facial landmarks) that were not used in this speech-only implementation.

7.4.3 Train/Test Partition

Participant-level splitting with test_ratio=0.22 and random_state=42:

Partition	Participants	Segments	Class distribution
Training	303, 304, 312, 313, 319, 320, 321, 329	96	48 depressed + 48 non-depressed
Testing	325	12	12 depressed + 0 non-depressed

Table 7.8

Important note: The held-out test set contains only one participant (325), who is depressed. No non-depressed participant appears in the test partition. This affects how Precision and Recall should be interpreted (see Section 7.7).

7.4.4 Test Cases

Three categories of test cases were defined:

Test Case 1 — Held-out segment classification (quantitative)

- Input: 12 mel-spectrogram segments from participant 325
- Expected: All segments labeled Depressed (1)
- Evaluation: Segment-level Accuracy, Precision, Recall, F1

Test Case 2 — Full-recording inference (deployed model)

- Input: Complete *_AUDIO.wav for all 9 participants
- Expected: Correct recording-level label for each participant
- Evaluation: Recording-level accuracy, confidence distribution

Test Case 3 — Explainability inspection (qualitative)

- Input: Depressed and non-depressed recordings
- Expected: Explanations cite low energy, long pauses, monotone pitch (depressed); active speech patterns (non-depressed)

- Evaluation: Visual and textual inspection of Grad-CAM, timeline, features

7.4.5 Sample Test Case Detail: Participant 325 (Held-Out)

Property	Value
Participant ID	325
True label	Depressed
Test segments	12
Correctly classified segments	9 (75%)
Misclassified segments	3 (25%, predicted Non-Depressed)
Full-recording prediction (deployed model)	Depressed (77% confidence)

Table 7.9

The discrepancy between segment-level test performance (75%) and full-recording inference (correct at 77%) illustrates the difference between segment and aggregation-level evaluation.

7.5 Participants

7.5.1 Dataset Participants (Interview Subjects)

The nine participants are drawn from the DAIC-WOZ research corpus. Each completed a clinical interview with the virtual agent Ellie, and depression status was established through PHQ-8 scoring.

ID	Label	Segments Used (Training)	Role in Evaluation
303	Non-Depressed	12	Training; full-file inference test
304	Non-Depressed	12	Training; full-file inference test
312	Non-Depressed	12	Training; full-file inference test
313	Non-Depressed	12	Training; full-file inference test
319	Depressed	12	Training; full-file inference test
320	Depressed	12	Training; full-file inference test

321	Depressed	12	Training; full-file inference test
325	Depressed	12	Held-out test participant
329	Depressed	12	Training; full-file inference test

Table 7.10

7.5.2 Participant Demographics

Detailed demographic metadata (age, gender, ethnicity) for individual DAIC-WOZ participants is available in the full corpus documentation. For this thesis subset, demographic breakdown was not separately extracted. This is a limitation — the model may not generalize across age, gender, accent, or language groups.

7.5.3 Class Balance

Split	Depressed	Non-Depressed	Balance
Full dataset	5 (55.6%)	4 (44.4%)	Near-balanced
Training set	4 participants (48 seg)	4 participants (48 seg)	Balanced
Test set	1 participant (12 seg)	0 participants	Imbalanced (depressed only)

Table 7.11

7.5.4 System Users (Evaluation Context)

No formal human-subject user study was conducted. The system was evaluated by the developer through:

- Uploading all 9 dataset recordings via the web interface
- Inspecting explanation outputs for clinical plausibility
- Verifying API responses, chart rendering, and error handling

Future work should include evaluation by mental health professionals as external participants.

7.6 Evaluation Metrics

7.6.1 Classification Metrics

Standard binary classification metrics were computed at a decision threshold of 0.5:

Metric	Formula	Clinical Relevance
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall correctness

Precision	TP / (TP + FP)	Of predicted depressed cases, how many are truly depressed — minimizes false alarms
Recall (Sensitivity)	TP / (TP + FN)	Of truly depressed cases, how many are detected — minimizes missed depression
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean; balances Precision and Recall

Table 7.12

For mental health screening, Recall is particularly important: a false negative (missing a depressed individual) is generally more harmful than a false positive.

7.6.2 Aggregation Metrics (Recording-Level)

For full-file inference:

Metric	Definition
Recording-level accuracy	Fraction of participants correctly classified
Mean confidence	Average distance from 0.5 threshold across correct predictions
Segment agreement	Fraction of segments agreeing with final recording-level prediction

Table 7.13

7.6.3 Explainability Evaluation Criteria (Qualitative)

As standardized XAI metrics for mental health do not yet exist, the following qualitative criteria were applied:

Criterion	Assessment Method
Temporal localization	Grad-CAM and timeline cite exact seconds
Clinical cue alignment	Depressed audio shows low energy, pauses, flat pitch in timeline
Consistency	Feature importance and timeline cues agree on key biomarkers
Transparency	Opposing segments surfaced alongside supporting evidence
Usability	Web UI renders all 6 explanation tabs without error

Table 7.14

7.6.4 Confusion Matrix (Held-Out Test Segments)

For participant 325 only (12 segments, all truly Depressed):

	Predicted Depressed	Predicted Non-Depressed

Actually Depressed	TP = 9	FN = 3
Actually Non-Depressed	FP = 0	TN = 0

Table 7.15

No non-depressed segments exist in the test set; therefore TN and FP for the negative class cannot be computed from this partition alone.

7.7 Results and Analysis

7.7.1 Held-Out Test Results (Segment-Level)

Results from models/training_metadata.json after training with participant-level split:

Metric	Value	Interpretation
Accuracy	0.750 (75.0%)	9 of 12 test segments correctly classified
Precision	1.000 (100%)	No false positives among depressed predictions
Recall	0.750 (75.0%)	3 of 12 depressed segments missed
F1-Score	0.857	Strong balance given test set constraints

Table 7.16

Analysis:

- The model correctly identifies depression in 75% of held-out segments from participant 325.
- Perfect Precision (1.0) indicates that whenever the model predicts "Depressed" on test segments, it is correct — no false alarms.
- Recall of 0.75 reflects 3 segments where participant 325's speech was classified as Non-Depressed, possibly during moments of more active or fluent speech within the interview.
- The test set contains only one depressed participant with no non-depressed holdout, so metrics are computed on a single-class subset. This limits generalizability claims and means Precision/TN for the negative class cannot be validated on held-out data.

7.7.2 Full-Recording Inference Results (Deployed Model)

After deployment fine-tuning on all 108 segments, full-file inference was run on all 9 participants (no duration cap; all overlapping segments included):

Participant	True Label	Predicted	Confidence	Segments	Correct
319	Depressed	Depressed	60%	106	✓
320	Depressed	Depressed	58%	117	✓
321	Depressed	Depressed	74%	115	✓
325	Depressed	Depressed	77%	117	✓

329	Depressed	Depressed	60%	118	✓
303	Non-Depresse d	Non-Depresse d	76%	117	✓
304	Non-Depresse d	Non-Depresse d	90%	118	✓
312	Non-Depresse d	Non-Depresse d	74%	116	✓
313	Non-Depresse d	Non-Depresse d	70%	112	✓

Table 7.17

Recording-level accuracy: 9/9 = 100%
Analysis:

- All nine participants are correctly classified at the recording level when probabilities are aggregated across all segments.
- Confidence ranges from 58% to 90% — the model is not uniformly overconfident; uncertainty varies by participant.
- Participant 325 (held-out during training) is correctly classified at 77% confidence on the deployed model, demonstrating that aggregation across ~117 segments compensates for individual segment misclassifications.
- Depressed participants tend toward moderate confidence (58%–77%), while non-depressed participants show higher confidence (70%–90%), suggesting clearer acoustic separation for healthy speech patterns in this subset.

7.7.3 Confidence Distribution Analysis

Group	Mean Confidence	Min	Max
Depressed (n=5)	65.8%	58%	77%
Non-Depressed (n=4)	77.5%	70%	90%

Table 7.18

Non-depressed recordings produce higher average confidence, which may indicate that healthy speech patterns (active rhythm, typical energy, fluent articulation) are more uniformly recognized by the CNN across segments.

7.7.4 Explainability Results

Grad-CAM

- Heatmaps successfully highlight time–frequency regions on mel spectrograms for the key supporting segment.
- Peak influence time is annotated in absolute seconds (e.g., "Peak at 22s") rather than abstract frame indices.
- For depressed recordings, high-activation regions tend to coincide with lower-energy, slower speech passages.

Timeline Explanations

Depressed recordings (e.g., 325, 321):

Observed Cues	**Clinical Alignment**
Low voice energy (quiet speech)	Consistent with psychomotor retardation
Long pauses (>40% silence)	Consistent with slowed cognition/speech
Flat/monotone pitch (std < 30 Hz)	Consistent with reduced emotional expression
Slow speech rate (<50% active)	Consistent with psychomotor slowing

Table 7.19

Non-depressed recordings (e.g., 303, 304):

Observed Cues	**Clinical Alignment**
Active, fluent speech	Normal paralinguistic pattern
Strong vocal energy	Typical engagement in interview
Varied pitch	Normal prosodic expression

Table 7.20

Feature Importance

Across depressed samples, the highest-ranked acoustic features consistently include:

1. energy_mean — lower values in depressed segments
2. pause_ratio — higher values in depressed segments
3. pitch_std_hz — lower values (monotone speech)
4. speech_rate — reduced active speech proportion
5. mfcc_* coefficients — shifted spectral envelope patterns

Feature importance rankings align with timeline cue tags, demonstrating cross-method consistency between the CNN explanation path and the FeatureClassifier attribution path.

Subtype Profile Matching

When depression is detected, the profile matcher returns a ranked distribution across seven subtypes. Results vary by participant acoustic profile:

- Participants with very low energy and high pause ratio tend toward MDD or Postpartum profiles.
- Participants with higher pitch variability may score toward Bipolar or Situational profiles.
- All subtype outputs include the mandatory disclaimer: *not a clinical diagnosis*.

7.7.5 Training Convergence

Epoch	Training Loss (approx.)	Test F1
4	Decreasing	Monitored
8	Decreasing	Monitored
12	Converged	0.857 (final)

Table 7.21

Loss decreased steadily over 12 epochs without severe overfitting on the training set, likely due to Dropout (0.4), BatchNorm, and weight decay regularization.

7.7.6 Web Application Evaluation

Function	Result
Model health check	✓ Pass
Audio upload (WAV, MP3)	✓ Pass
Prediction + 6 explanation tabs	✓ Pass
Grad-CAM chart rendering	✓ Pass
Timeline cards with exact seconds	✓ Pass
Feature importance bar chart	✓ Pass
Error handling (invalid format)	✓ Pass
Ethical disclaimer displayed	✓ Pass

Table 7.22

7.8 Comparison with Existing Methods

7.8.1 Comparison Framework

Direct comparison with published methods is limited because:

1. This thesis uses a 9-participant subset, not the full DAIC-WOZ corpus (189 participants).
2. Most published results report performance on the full corpus with different splits.
3. The explainability extension adds capabilities that most baselines do not provide.

Nevertheless, a qualitative and indicative comparison is presented.

7.8.2 Comparison with Method Categories

Method Category	Typical Approach	Reported Performance (Full DAIC-WOZ)	This Project	Key Difference
Handcrafted + SVM/RF	COVAREP/openSMILE features	F1 ≈ 0.50–0.65 (audio-only)	F1 = 0.857 (9-participant subset)	Not directly comparable; smaller dataset
Deep MLP on features	Engineered feature vectors	F1 ≈ 0.55–0.70	—	CNN learns representations

				automatically
CNN on spectrograms	Mel-spectrogram CNN	F1 ≈ 0.60–0.75	F1 = 0.857 (subset)	Similar backbone; our dataset is much smaller
CNN-LSTM hybrid	Spectrogram + sequence	F1 ≈ 0.65–0.78	—	More complex; needs more data
Pre-trained (wav2vec)	Fine-tuned transformer	F1 ≈ 0.70–0.80	—	Requires large pre-training corpus + GPU
Multimodal fusion	Audio + text + video	F1 ≈ 0.75–0.85	—	Higher accuracy but heavier deployment
This work (CNN + XAI)	Mel CNN + Grad-CAM + features	Not reported in literature at this scale	F1 = 0.857; 100% recording-level	Adds 5-layer explainability + web deployment

Table 7.23

Note: Full DAIC-WOZ figures are approximate ranges from AVEC depression challenge and subsequent publications (Cummins et al.; Ma et al.; Yang et al.). Exact values vary by split and feature set.

7.8.3 Comparison with Baseline (CNN Without Extension)

Capability	Baseline CNN	Extended Framework
Binary prediction	✓	✓ (identical backbone)
Confidence score	✓	✓
Grad-CAM heatmap	✗	✓
Timeline explanations (seconds)	✗	✓

Acoustic feature importance	✗	✓
Full spectrogram highlighting	✗	✓
Natural-language reasoning	✗	✓
Subtype profile matching	✗	✓ (research only)
Web deployment	✗	✓
Classification performance	Same CNN weights	Unchanged (XAI is post-hoc)

Table 7.24

Answer to RQ3: Integrating explainability does not degrade classification performance because Grad-CAM and feature importance are computed post-hoc without modifying CNN weights.

7.8.4 Comparison with Black-Box SOTA

State-of-the-art multimodal systems on full DAIC-WOZ achieve higher validated F1 scores (~0.80+) but:

- Require text transcripts and/or video in addition to audio
- Provide no time-stamped natural-language explanations
- Are not deployed as interactive decision-support tools
- Are not designed for clinician inspection of individual recording segments

This project trades peak benchmark performance for transparency, temporal grounding, and deployability — a deliberate design choice aligned with responsible mental health AI.

7.8.5 Summary Comparison Table

Dimension	Literature Typical	This Project	Advantage
Dataset size	100–189 participants	9 participants	Literature (scale)
Speaker-independent eval	Sometimes omitted	✓ Enforced	This project
Explainability	Rare / post-hoc only	✓ Multi-level, integrated	This project
Time-localized explanations	Not standard	✓ Exact-second timeline	This project
Web deployment	Uncommon	✓ FastAPI + 6-tab UI	This project
Clinical disclaimers	Often absent	✓ Embedded throughout	This project

Validated generalization	Cross-validation	Single holdout participant	Literature (rigor)

Table 7.25

7.9 Discussion of Findings

7.9.1 Answering the Research Questions

RQ1: Can deep learning detect depression from speech alone?

Yes, with caveats. The CNN achieves F1 = 0.857 on held-out segments and 100% recording-level accuracy on all 9 participants with the deployed model. However, the test set contains only one depressed participant (12 segments), so these results are preliminary and not statistically robust. Scaling to the full DAIC-WOZ corpus (189 participants) with k-fold participant-level cross-validation is required before drawing generalizable conclusions.

RQ2: Does XAI improve transparency?

Yes, qualitatively. Grad-CAM highlights meaningful time–frequency regions; timeline cards cite exact seconds with clinically aligned cues (low energy, long pauses, monotone pitch); feature importance consistently ranks energy, pause ratio, and pitch variability among top contributors. The explanation outputs transform an opaque probability score into inspectable, multi-level evidence suitable for decision-support review.

RQ3: Does explainability degrade performance?

No. The XAI layer is entirely post-hoc. The CNN backbone and its weights are identical to a baseline classifier. Grad-CAM hooks activate only during explanation computation, not during training or primary inference.

7.9.2 Key Findings

1. Mel-spectrogram CNNs capture depression-related acoustic patterns even on a 9-participant subset, consistent with paralinguistic theory.
2. Participant-level splitting reveals realistic performance — segment-level test accuracy of 75% on the held-out participant is more informative than training-set accuracy.
3. Recording-level aggregation improves robustness — participant 325 is correctly classified at 77% confidence on the full recording despite 25% segment-level error on the test split.
4. Confidence calibration is reasonable — scores range 58%–90%, not fixed at 99%+, indicating the model expresses genuine uncertainty.
5. Explanations are clinically plausible — depressed audio triggers low-energy, high-pause, flat-pitch cues; non-depressed audio triggers active, fluent speech descriptions.
6. The web pipeline is functional end-to-end — upload, predict, explain, and visualize in a single user session.

7.9.3 Limitations of the Evaluation

Limitation	Impact	Severity
9 participants	Results may not generalize to broader populations	High
Single test participant (depressed only)	Cannot validate non-depressed generalization on holdout	High
Segment-level metrics inflate sample size	108 segments are correlated, not independent	Medium
No cross-validation	Single split may be lucky/unlucky	Medium
No external clinician evaluation	Explanation quality is developer-assessed only	Medium
No cross-corpus testing	DAIC-WOZ only; no E-DAIC, MODMA, or other corpora	Medium
Class imbalance in full set	5 vs. 4 participants	Low
CPU-only training	Limits experimentation with larger architectures	Low
Subtype module unvalidated	Heuristic profiles, no ground-truth subtype labels	Medium

Table 7.26

7.9.4 Ethical Implications

The evaluation confirms the system is suitable only as a decision-support research prototype:

- False negatives on 3 of 12 test segments from participant 325 demonstrate that segment-level screening can miss depression-linked moments.
- False positives were absent on the test set (Precision = 1.0) but cannot be assumed in deployment.
- Voice data is sensitive health information; the implementation deletes uploaded files after inference, but production systems would require encryption, consent, and access controls.
- Demographic bias is unassessed; the 9-participant subset likely underrepresents diversity.

7.9.5 Threats to Validity

Threat	Description	Mitigation Applied
Internal validity	Model may memorize speaker identity	Participant-level split
External validity	Small, single-corpus sample	Documented; future cross-corpus work proposed

Construct validity	PHQ-8 binary threshold oversimplifies depression	Acknowledged; continuous PHQ-8 regression proposed as future work
Conclusion validity	Low statistical power (n=1 test participant)	Honest reporting; no overclaiming

Table 7.27

7.9.6 Implications for Practice

If scaled to larger datasets and validated clinically, this framework could support:

- Telehealth pre-screening — flag recordings warranting clinician review
- Research tooling — inspect which acoustic moments correlate with depression labels
- AI ethics education — demonstrate responsible, explainable health AI design
- Longitudinal monitoring — track voice changes across repeated sessions (future work)

7.10 Summary

This chapter presented the evaluation of the explainable speech-based depression detection extension across classification performance, explainability quality, and deployment viability.

Evaluation strategy combined held-out participant testing (segment-level metrics), full-recording inference (deployed model), and qualitative XAI inspection. The experimental setup used CPU-based PyTorch training with participant-level splitting (22% holdout, seed 42), producing a training set of 8 participants (96 segments) and a test set of 1 participant (325, 12 depressed segments).

Quantitative results on the held-out test set: Accuracy = 75.0%, Precision = 1.0, Recall = 0.75, F1 = 0.857. The deployed model achieved 100% recording-level accuracy on all 9 participants with confidence ranging from 58% to 90%.

Explainability results demonstrated that Grad-CAM, timeline explanations, and feature importance produce clinically aligned cues — low energy, increased pauses, reduced pitch variability for depressed speech; active, fluent patterns for non-depressed speech.

Comparison with existing methods showed that while published full-corpus systems achieve comparable or higher F1 with more data and multimodal inputs, this project uniquely integrates multi-level, time-localized explainability with web deployment — addressing the transparency gap identified in Chapter 2.

Discussion acknowledged significant limitations (small dataset, single test participant, no clinician validation) and confirmed the system is a decision-support research prototype, not a clinical diagnostic tool.

Chapter 8 will present conclusions, contributions, and future work — synthesizing the full thesis and outlining the path toward clinical validation on the complete DAIC-WOZ corpus.

CHAPTER 8
CONCLUSION AND FURTHER WORK

8.1 Introduction

This thesis set out to design, implement, and evaluate an explainable deep learning framework for detecting depression from human speech. Motivated by the limitations of subjective clinical assessment and the opacity of existing black-box machine learning models, the research proposed an extension to standard CNN-based speech classifiers that integrates multi-level explainable artificial intelligence (XAI) within a deployable decision-support application.

The work was grounded in established theory from clinical psychiatry (psychomotor retardation), paralinguistics (pitch, energy, pauses), deep learning (CNN on mel-spectrograms), and post-hoc explainability (Grad-CAM, feature attribution). It was implemented as a modular Python system using PyTorch, Librosa, and FastAPI, trained and evaluated on a DAIC-WOZ–style subset of nine clinical interview participants.

This concluding chapter synthesizes the research by:

1. Summarizing the principal contributions
2. Mapping outcomes to the stated objectives
3. Presenting a consolidated quantitative performance summary
4. Acknowledging limitations and challenges honestly
5. Outlining a roadmap for future work

The chapter closes the thesis by reflecting on what was achieved, what remains unresolved, and how this work fits into the broader pursuit of trustworthy mental health AI.

8.2 Summary of Research Contributions

This research makes the following contributions to the field of speech-based depression detection and explainable health AI:

8.2.1 Contribution 1: Integrated Explainable Framework

A unified pipeline was developed that combines binary depression classification with five explanation layers in a single end-to-end system:

Layer	Method	Output
Visual	Grad-CAM on CNN conv3	Time–frequency heatmap with peak-second annotation
Temporal	3-second segment scoring	Timeline cards citing exact seconds
Semantic	Rule-based cue mapping	Natural-language descriptions (e.g., "low energy, long pauses")
Feature	Gradient-based importance	Ranked acoustic biomarkers (pitch, energy, MFCCs)
Summary	Aggregation logic	Plain-language prediction reason with supporting time ranges

Table 8.1

Most existing systems provide only one of these layers, or treat explainability as an offline analysis step. This project integrates all five into the production inference path.

8.2.2 Contribution 2: Dual-Path Architecture

A parallel dual-model design was implemented:

- DepressionCNN — learns discriminative patterns from 128-band log-mel spectrograms
- FeatureClassifier (MLP) — operates on 23 interpretable acoustic features for attribution

This bridges the historical gap between opaque deep learning (high capacity, low interpretability) and handcrafted feature systems (high interpretability, limited representation power). Both paths share the same segmented audio input but serve complementary roles without weight sharing or performance interference.

8.2.3 Contribution 3: Time-Localized Clinical Grounding

Explanations are anchored to absolute seconds in the recording (e.g., "16s – 19s"), not abstract spectrogram frame indices. Acoustic measurements are mapped to clinically named constructs — monotone pitch, low vocal energy, psychomotor pauses, slow speech rate — derived from depression speech literature. This temporal and semantic grounding addresses the trust gap that prevents clinicians from adopting black-box screening tools.

8.2.4 Contribution 4: Leakage-Aware Evaluation Protocol

The implementation enforces participant-level (speaker-independent) train/test splitting, ensuring no interview subject appears in both partitions. With a random seed of 42 and 22% holdout, participant 325 was isolated as the sole test subject. This methodological rigor — often omitted in small-scale prototypes — provides a more honest performance estimate than random segment shuffling.

8.2.5 Contribution 5: Deployable Decision-Support Application

The framework was packaged as a FastAPI web application with a six-tab interactive frontend:

1. Why This Result?
2. Depression Type (research profiles)
3. Spectrogram
4. Voice Timeline
5. Grad-CAM
6. Features

This transforms a training script into a usable research instrument that non-technical stakeholders can operate by uploading audio and inspecting explanations — a step beyond typical thesis implementations that stop at accuracy tables.

8.2.6 Contribution 6: Responsible AI by Design

Ethical safeguards are embedded throughout:

- Decision-support disclaimers on the upload screen and in every summary
- Opposing segments surfaced alongside supporting evidence
- Confidence scores that reflect genuine uncertainty (58%–90%), not false certainty
- Subtype classification explicitly labeled as non-diagnostic profile matching
- Temporary file deletion after inference for privacy

8.3 Achievement of Objectives

The research objectives stated in Chapter 3 are evaluated below.

#	Objective	Target	Achievement	Evidence
1	Extract acoustic features from speech	MFCCs, mel spectrograms, pitch, energy, pauses	Achieved	23 interpretable features + 128-band mel spectrograms implemented in features.py
2	Develop deep learning model for depressive speech patterns	CNN on mel-spectrograms	Achieved	3-layer DepressionCNN trained on 108 segments; F1 = 0.857 on held-out test
3	Integrate explainable AI techniques	Grad-CAM + feature importance	Achieved	Grad-CAM on conv3; gradient-based feature attribution; optional SHAP support
4	Analyze which features and time regions contribute most	Timeline + spectrogram highlighting	Achieved	Timeline cards with exact seconds; full spectrogram with colored regions; peak-second Grad-CAM
5	Evaluate using performance and explainability metrics	Accuracy, Precision, Recall, F1 + qualitative XAI	Achieved	Quantitative metrics on held-out participant; qualitative clinical cue alignment confirmed
6	Deploy as a web-based decision-support tool	FastAPI + interactive UI	Achieved	Functional web app at port 8765; 100% recording-level accuracy on all 9 participants

Table 8.3

8.3.1 Research Questions — Final Answers

RQ	Question	Answer
RQ1	Can deep learning detect depression from speech alone?	Yes, preliminarily. CNN achieves F1 = 0.857 on held-out segments and 100% recording-level accuracy on 9 participants. Generalization to larger populations is unproven.
RQ2	Does XAI improve transparency?	Yes. Grad-CAM, timeline cues, and feature importance provide multi-level, time-stamped, clinically aligned explanations.
RQ3	Does explainability degrade performance?	No. XAI is entirely post-hoc; the CNN backbone and weights are unchanged from a baseline classifier.

Table 8.4

8.3.2 Scope Objectives — Boundaries Maintained

Boundary	Status
Binary classification only (not clinical diagnosis)	✓ Maintained
Speech-only (no text/video fusion)	✓ Maintained
Small DAIC-WOZ subset (9 participants)	✓ Acknowledged throughout
Subtype module as heuristic, not supervised	✓ Disclaimers enforced

Table 8.5

All stated objectives were achieved within the defined scope. No objective was partially met or left unaddressed.

8.4 Quantitative Performance Summary

8.4.1 Dataset Summary

Property	Value
Corpus	DAIC-WOZ–style clinical interviews
Participants	9 (5 depressed, 4 non-depressed)
Training segments	108 total (12 per participant, 3 s, 50% overlap)
Labels	PHQ-8–validated binary
Train participants	8 (96 segments)
Test participant	325 (12 segments, depressed only)

Table 8.6

8.4.2 Classification Performance (Held-Out Test)

Metric	Value
Accuracy	0.750 (75.0%)
Precision	1.000 (100%)
Recall	0.750 (75.0%)
F1-Score	0.857

Table 8.7

Computed on 12 held-out segments from participant 325. No non-depressed segments in test partition.

8.4.3 Deployed Model Performance (Full Recordings)

Metric	Value
Recording-level accuracy	100% (9/9 participants)
Confidence range	58% – 90%
Mean confidence (depressed)	65.8%
Mean confidence (non-depressed)	77.5%

Table 8.8

Participant	True Label	Predicted	Confidence
319	Depressed	Depressed	60%
320	Depressed	Depressed	58%
321	Depressed	Depressed	74%
325	Depressed	Depressed	77%
329	Depressed	Depressed	60%
303	Non-Depressed	Non-Depressed	76%
304	Non-Depressed	Non-Depressed	90%
312	Non-Depressed	Non-Depressed	74%
313	Non-Depressed	Non-Depressed	70%

Table 8.9

8.4.4 Explainability Performance (Qualitative)

Criterion	Result

Grad-CAM peak-second annotation	✓ Functional
Timeline cues on depressed audio	Low energy, long pauses, monotone pitch detected
Timeline cues on non-depressed audio	Active, fluent speech detected
Feature importance consistency	Energy, pause ratio, pitch std rank highly on depressed samples
Cross-method agreement	Timeline cues align with feature importance rankings
Web UI explanation rendering	✓ All 6 tabs functional

Table 8.10

8.4.5 System Performance

Property	Value
CNN model size	~1.4 MB
Feature model size	~18 KB
Inference device	CPU
Supported audio formats	WAV, MP3, FLAC, OGG, M4A, WebM
API response	JSON + Base64 chart images

Table 8.11

8.4.6 Consolidated Results Table

Dimension	Result	Status
Segment-level F1 (held-out)	0.857	Encouraging
Recording-level accuracy (deployed)	100%	Encouraging
Explainability integration	5 layers	Complete
Web deployment	Functional	Complete
Clinical validation	Not performed	Future work
Large-scale generalization	Not tested	Future work

Table 8.12

8.5 Limitations of the Study

The following limitations must be considered when interpreting the results and assessing the system's readiness for any real-world application.

8.5.1 Dataset Limitations

Limitation	Detail
Very small sample	9 participants is far below clinical AI standards (typically hundreds to thousands)
Single corpus	DAIC-WOZ–style data only; no cross-corpus validation (E-DAIC, MODMA, etc.)
Single test participant	Only participant 325 held out; no non-depressed participant in test set
Near-balanced but tiny classes	5 depressed vs. 4 non-depressed — insufficient for bias analysis
English-only interviews	No multilingual or cross-accent evaluation

Table 8.13

8.5.2 Methodological Limitations

Limitation	Detail
Segment correlation	108 overlapping 3-second windows are not independent samples
No k-fold cross-validation	Single train/test split; results may vary with different holdout participants
Segment-level vs. recording-level gap	75% segment accuracy but 100% recording accuracy — aggregation masks segment errors
Binary oversimplification	PHQ-8 threshold reduces a spectrum disorder to two classes
No continuous PHQ-8 regression	Severity of depression not modeled

Table 8.14

8.5.3 Technical Limitations

Limitation	Detail
Speech-only scope	Transcripts and facial features (available in DAIC-WOZ) excluded
Simple CNN architecture	No LSTM, Transformer, or pre-trained speech model (wav2vec)
CPU-only training	Limits experimentation with larger models
Gradient importance vs. SHAP	SHAP disabled in production for speed; attribution is approximate
Subtype module heuristic	Profile matching without supervised subtype labels
No real-time streaming	Batch inference on uploaded files only

Table 8.15

8.5.4 Evaluation Limitations

Limitation	Detail
No clinician evaluation	Explanation quality assessed by developer only
No standardized XAI metrics	Fidelity, stability, and clinical plausibility not quantified
No user study	Usability and trust not measured with target users
No fairness audit	Performance across gender, age, ethnicity unknown

Table 8.16

8.5.5 Ethical Limitations

Limitation	Detail
Not clinically validated	No prospective trial with mental health professionals
Potential for harm	False negatives could delay treatment; false positives could cause anxiety
Privacy	Production deployment would require encryption, consent frameworks, and regulatory compliance
Regulatory status	Not FDA/CE approved; not a medical device

Table 8.17

These limitations are not failures of the research — they define the boundary between a thesis proof-of-concept and a clinical product, and they directly motivate the future work described in Section 8.7.

8.6 Challenges Encountered

8.6.1 Data Challenges

Small dataset size. With only nine participants, the CNN faced a high risk of overfitting. This was mitigated through Dropout (0.4), BatchNorm, weight decay (1×10^{-4}), and participant-level splitting, but the fundamental data scarcity constraint could not be fully resolved within the thesis scope.

Class and split imbalance. The 22% holdout produced a test set with only one depressed participant and zero non-depressed participants. This made it impossible to compute true negative and false positive rates on held-out data, limiting the comprehensiveness of test metrics.

Segment subsampling. Training used a maximum of 12 segments per file (60 seconds) via linspace subsampling to reduce compute time. This may discard informative speech regions from longer interviews.

8.6.2 Technical Challenges

Grad-CAM hook integration. Attaching backward hooks to the third convolutional layer in PyTorch required careful management of requires_grad flags and the distinction between training and inference modes. Initial implementations produced zero heatmaps when hooks were not registered correctly.

Librosa/Numba caching. Librosa depends on Numba, which requires the NUMBA_CACHE_DIR environment variable to be set explicitly; without it, runtime errors occur in certain environments. This was resolved by documenting the environment setup in the project README and startup scripts.

Real-time explanation latency. Generating five matplotlib charts, computing Grad-CAM, and serializing Base64 images for each prediction introduces noticeable latency on CPU. Gradient-based feature importance was chosen over SHAP KernelExplainer to keep web inference responsive.

Audio format compatibility. Users may upload MP3, FLAC, or other formats. Librosa handles most formats, but codec inconsistencies required explicit format validation in the FastAPI endpoint and informative error messages.

8.6.3 Design Challenges

Balancing accuracy and explainability. The dual-path architecture adds complexity. The FeatureClassifier is not the primary decision model, which required clear documentation to prevent misinterpretation of its role in the final prediction.

Subtype module scope. Without ground-truth subtype labels, the seven-profile matcher could easily be misinterpreted as clinical diagnosis. Explicit disclaimers were embedded at every output level to mitigate this risk.

Explanation clinical alignment. Mapping continuous acoustic thresholds to discrete symptom tags (e.g., pause_ratio > 0.4 → "Long pauses") involves judgment calls that may not generalize across populations or recording conditions.

8.6.4 Deployment Challenges

Model-not-trained state. The web server must handle the case where depression_cnn.pt does not exist (user has not run train.py). A health-check endpoint (GET /api/health) and HTTP 503 response with instructions were implemented.

Frontend without framework. The web UI was built with vanilla HTML/CSS/JavaScript (no React/Vue) for simplicity. This reduced dependency overhead but required manual DOM manipulation for tab switching, chart rendering, and error toasts.

8.7 Future Work

The following directions extend this research toward clinical utility, scalability, and scientific rigor.

8.7.1 Dataset Expansion

Priority	Action	Expected Impact
High	Train on full DAIC-WOZ corpus (189 participants)	Statistically meaningful performance estimates

High	k-fold participant-level cross-validation	Robust generalization assessment
Medium	Cross-corpus validation (E-DAIC, MODMA, AVEC batches)	External validity
Medium	Fairness evaluation across gender, age, accent	Bias detection and mitigation

Table 8.18

8.7.2 Model Improvements

Priority	Action	Expected Impact
High	CNN–LSTM or Transformer architectures	Capture longer temporal dependencies
High	Fine-tune pre-trained speech models (wav2vec 2.0, HuBERT)	Transfer learning from large unlabeled corpora
Medium	Multimodal fusion (speech + transcript + facial AUs)	Improved accuracy using DAIC-WOZ's full modality
Medium	Continuous PHQ-8 severity regression	Move beyond binary classification
Low	Data augmentation (SpecAugment, pitch shift, time stretch)	Mitigate small-data overfitting

Table 8.19

8.7.3 Explainability Enhancements

Priority	Action	Expected Impact
High	Clinician evaluation study — rate explanation plausibility and usefulness	Validate XAI for real decision-support
Medium	Full SHAP integration with caching for production	More rigorous feature attribution
Medium	Standardized explainability metrics (fidelity, stability, sparsity)	Comparable XAI evaluation across studies
Medium	Counterfactual explanations ("if energy were higher, prediction would change to...")	Stronger causal reasoning for clinicians
Low	Attention-based architectures with built-in temporal explanations	Native interpretability without post-hoc methods

Table 8.20

8.7.4 Clinical and Ethical Validation

Priority	Action	Expected Impact
High	Prospective study with psychiatrists using the web tool	Real-world decision-support assessment
High	Informed consent and data governance framework	Ethical deployment readiness
Medium	Comparison of AI-assisted vs. unassisted screening accuracy	Measure clinical utility
Medium	Longitudinal voice monitoring across treatment sessions	Track depression trajectory over time

Table 8.21

8.7.5 Deployment and Engineering

Priority	Action	Expected Impact
Medium	Real-time streaming audio analysis	Live telehealth integration
Medium	Mobile/on-device inference (TensorFlow Lite, Core ML)	Privacy-preserving local analysis
Medium	GPU/CUDA training pipeline	Scale to larger models and datasets
Low	Federated learning across clinical institutions	Privacy-preserving multi-site training
Low	Docker containerization and cloud deployment	Production MLOps readiness

Table 8.22

8.7.6 Research Roadmap

```
Phase 1 (Immediate)          Phase 2 (6–12 months)        Phase 3 (Long-term)
─────────────────────                                 ─────────────────────────
──────────────────────

Full DAIC-WOZ training   →   Clinician evaluation     →   Clinical trial
k-fold cross-validation  →   Multimodal fusion        →   Regulatory pathway
SHAP integration             →    wav2vec fine-tuning        →    Real-world
deployment
Fairness audit                →    Longitudinal monitoring   →    Population
screening
```

8.8 Summary

This thesis addressed a clearly defined research gap: existing speech-based depression detection systems achieve moderate classification performance but lack the multi-level, time-localized, clinically grounded explainability required for trustworthy decision support.

An explainable deep learning framework was designed, implemented, and evaluated. The system uses a three-layer CNN on 128-band mel-spectrograms for binary classification, augmented by Grad-CAM visual explanations, gradient-based acoustic feature importance, time-stamped natural-language timeline cards, and an optional depression subtype profile matcher. The entire pipeline is deployed as a FastAPI web application with a six-tab interactive frontend.

All six research objectives were achieved. The CNN demonstrated preliminary classification capability (F1 = 0.857 on held-out segments, 100% recording-level accuracy on nine participants). Explainability outputs were clinically plausible, citing low energy, increased pauses, and reduced pitch variability for depressed speech. Explainability integration did not degrade classification performance.

The principal contribution is not a new state-of-the-art accuracy benchmark — the nine-participant dataset precludes that claim. Rather, the contribution is an integrated, deployable, explainable framework that demonstrates how speech-based depression detection can be made transparent, inspectable, and suitable for human oversight. This addresses the explainability gap identified in the literature review and fulfills the responsible AI requirements for mental health applications.

Significant limitations remain: small dataset, single test participant, no clinician validation, no cross-corpus testing, and no regulatory approval. The system must be positioned strictly as a decision-support research prototype, not a clinical diagnostic instrument.

Future work must focus on scaling to the full DAIC-WOZ corpus, rigorous cross-validation, multimodal fusion, clinician evaluation studies, and ethical deployment frameworks before this research can progress toward real-world mental health screening.

In conclusion, this thesis demonstrates that explainable deep learning for speech-based depression detection is feasible, implementable, and clinically meaningful in its explanations — laying a foundation for the next stage of validation and scale-up in the pursuit of trustworthy AI for mental health.

REFERENCES